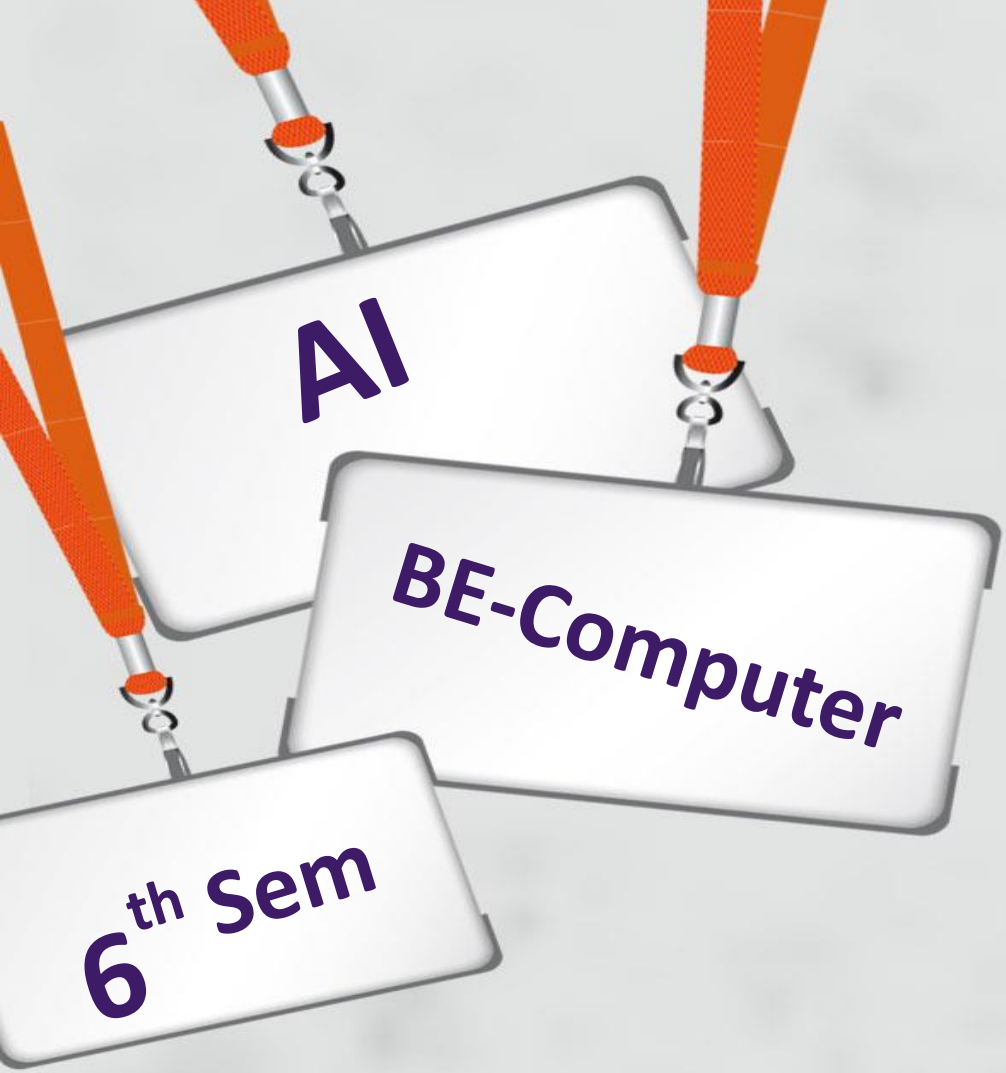


Chapter 6

Machine learning

Learning

- Learning is one of those everyday terms which is broadly and vaguely used in the English language
  - **Learning is making useful changes in our minds**
  - **Learning is constructing or modifying representations of what is being experienced**
  - **Learning is the phenomenon of knowledge acquisition in the absence of explicit programming**

Herbert Simon, 1983

> *Learning denotes changes in the system that are adaptive in the sense that they enable the system to do the same task or tasks drawn from the same population more efficiently and more effectively next time.*

- Learning involves the recognition of patterns in data or experience and then using that information to improve performance on another task.

Learning

- Learning involves 3 factors:

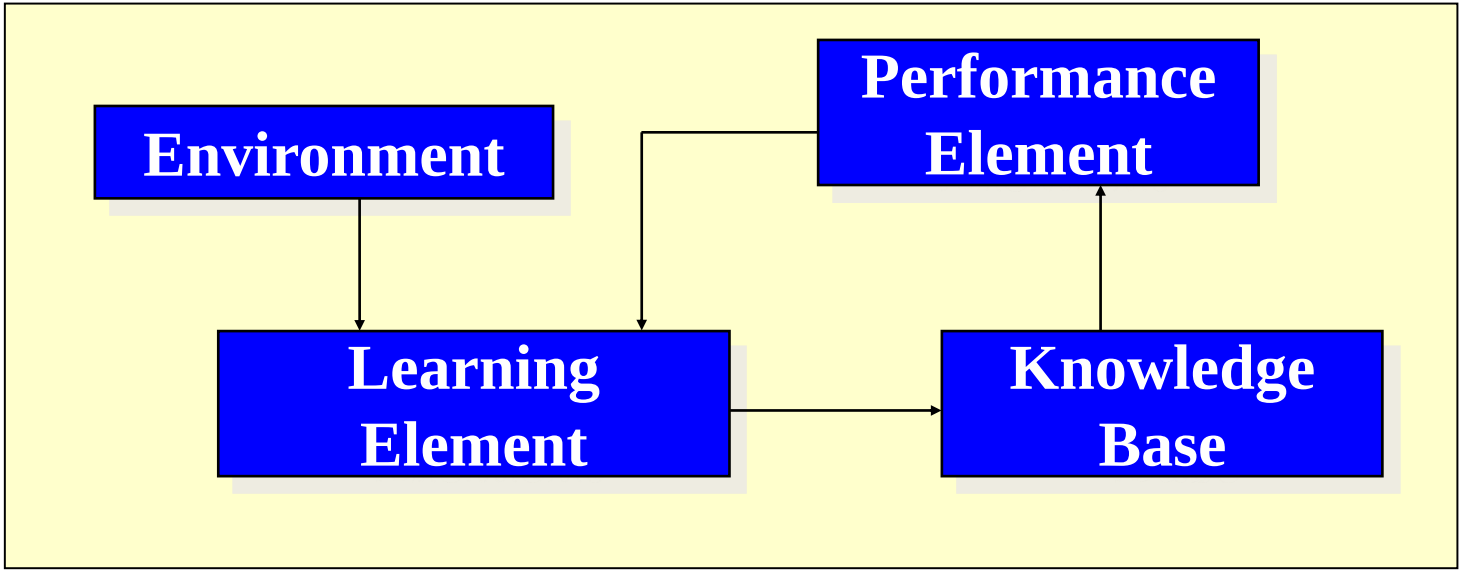
changes	**Learning changes the learner: for machine learning the problem is determining the nature of these changes and how to best represent them**
generalization	**Learning leads to generalization: performance must improve not only on the same task but on similar tasks**
improvement	**Learning leads to improvements: machine learning must address the possibility that changes may degrade performance and find ways to prevent it.**

Learning Methods

- There are two different kinds of information processing which must be considered in a machine learning system
  - ***Inductive learning*** is concerned with determining general patterns, organizational schemes, rules, and laws from raw data, experience or examples.
  - ***Deductive learning*** is concerned with determination of specific facts using general rules or the determination of new general rules from old general rules.

Learning Framework

- There are four major components in a learning system:



Learning Framework: **The Environment**

- The environment refers the nature and quality of information given to the learning element

- The nature of information depends on its level
  - high level information is abstract, it deals with a broad class of problems
  - low level information is detailed, it deals with a single problem.

- The quality of information involves
  - noise free
  - reliable
  - ordered

Learning Framework: **Learning Elements**

- Four learning situations
  - Rote Learning
    - environment provides information at the required level
  - Learning by being told
    - information is too abstract, the learning element must hypothesize missing data
  - Learning by example
    - information is too specific, the learning element must hypothesize more general rules
  - Learning by analogy
    - information provided is relevant only to an analogous task, the learning element must discover the analogy

Learning Framework: **Knowledge Base**

- Expressive
  - the representation contains the relevant knowledge in an easy to get to fashion
- Modifiable
  - it must be easy to change the data in the knowledge base
- Extendibility
  - the knowledge base must contain meta-knowledge (knowledge on how the data base is structured) so the system can change its structure

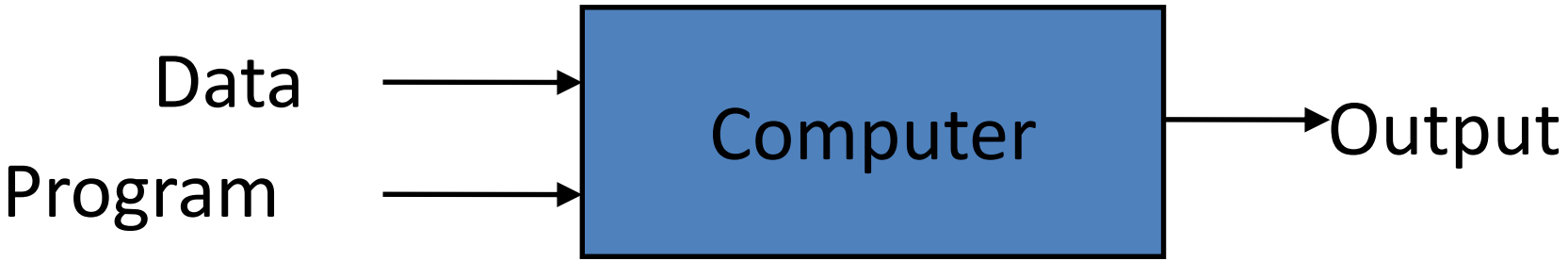
Learning Framework: **Performance Element**

- Complexity
  - for learning, the simplest task is classification based on a single rule while the most complex task requires the application of multiple rules in sequence
- Feedback
  - the performance element must send information to the learning system to be used to evaluate the overall performance
- Transparency
  - the learning element should have access to all the internal actions of the performance element

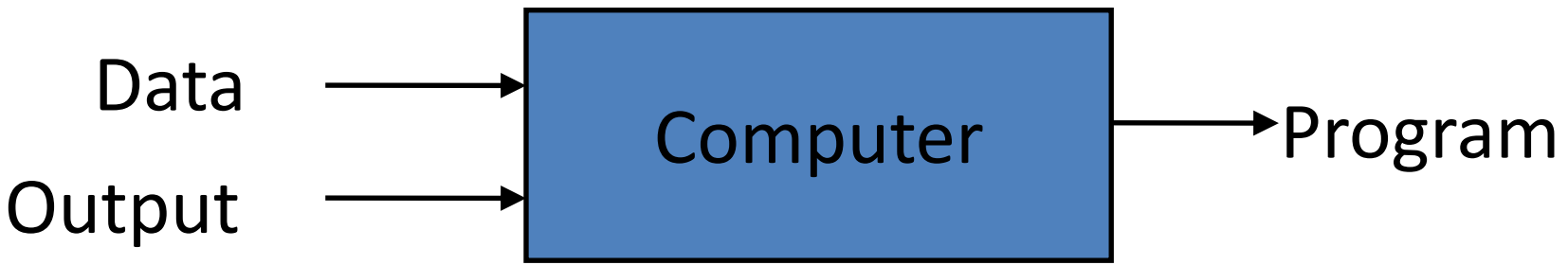
Machine Learning

- Machine learning is a type of artificial intelligence (AI) that provides computers with the ability to learn without being explicitly programmed. Machine learning focuses on the development of computer programs that can change when exposed to new data.
- The process of machine learning is similar to that of data mining. Both systems search through data to look for patterns. However, instead of extracting data for human comprehension -- as is the case in data mining applications -- machine learning uses that data to detect patterns in data and adjust program actions accordingly.

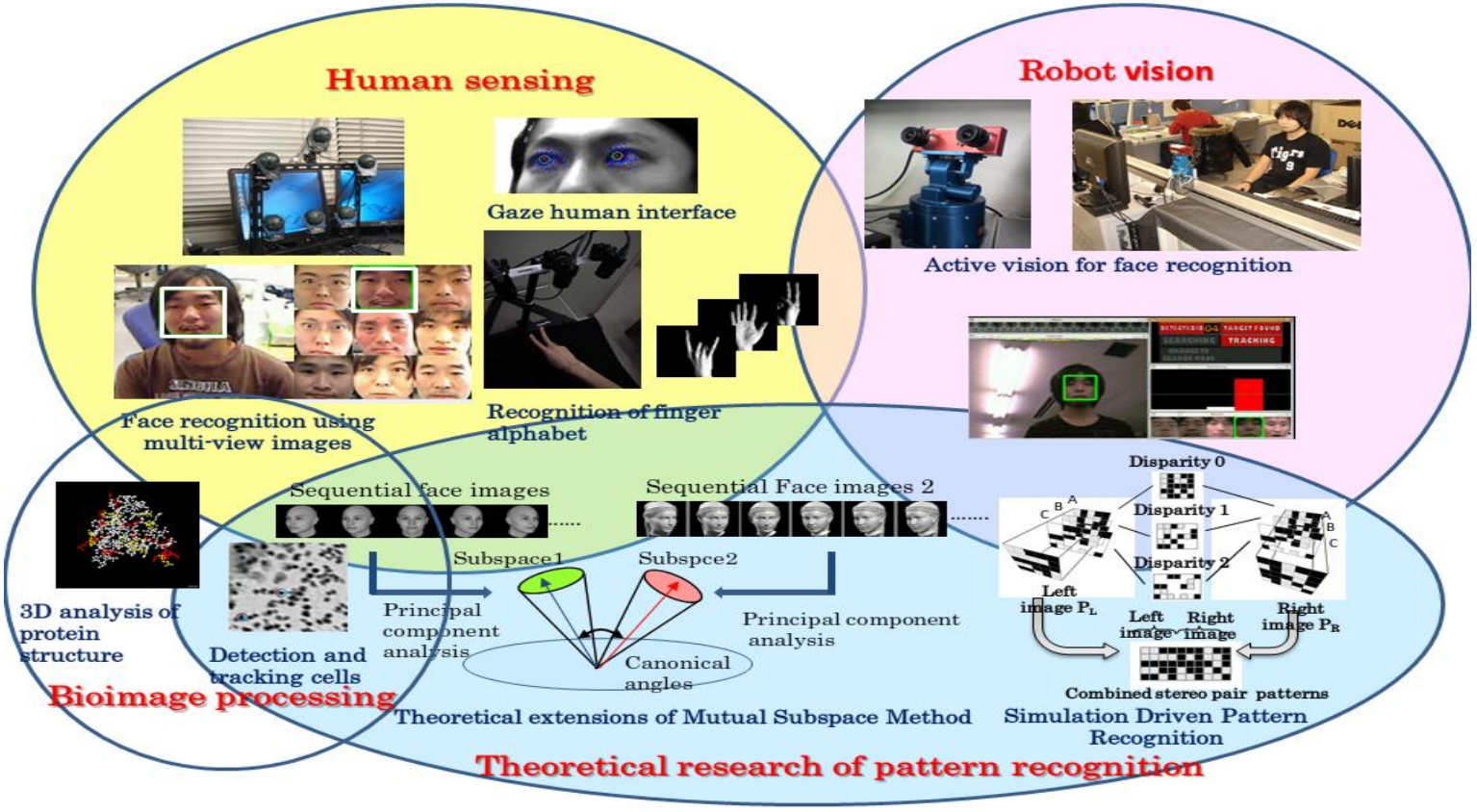
Traditional Programming



Machine Learning



Applications of ML



Examples of Machine Learning Problems

- Some of the examples of problems where Machine Learning can be used are:
  - **optical character recognition:** categorize images of handwritten characters by the letters represented
  - **face detection:** find faces in images (or indicate if a face is present)
  - **spam filtering:** identify email messages as spam or non-spam
  - **topic spotting:** categorize news articles (say) as to whether they are about politics, sports, entertainment, etc.
  - **spoken language understanding:** within the context of a limited domain, determine the meaning of something uttered by a speaker to the extent that it can be classified into one of a fixed set of categories

Explanation Based Learning (EBL)

- Humans appear to learn quite a lot from one example.
- Uses existing declarative domain knowledge to "explain" individual examples and uses this explanation to drive a knowledge-based generalization of the example.

- Produces a description of a concept that enables instances of the concept to be recognized efficiently.
- Capable of **learning a very general concept from very few training examples.**

Explanation Based Learning (EBL)

4 basic steps of EBL

- **A training example**
  what the learning sees in the world.
- **A goal concept**
  a high level description of what the program is supposed to learn.
- **A operational criterion**
  a description of which concepts are usable.
- **A domain theory**
  a set of rules that describe relationships between objects and actions in a domain.

Advantages/Disadvantages of EBL

- Advantages
  - needs just a small number of training examples

- Disadvantages
  - loss of coverage (EBL grammar is a subset of the original grammar)

Supervised Learning

- **Supervised learning** is where you have input variables (X) and an output variable (Y) and you use an algorithm to learn the mapping function from the input to the output.

  The goal is to approximate the mapping function so well that when you have new input data (X) that you can predict the output variables (Y) for that data.

  $$Y = f(X)$$

- The term supervised learning refers to the fact that we use a data set in which the correct answers were given. The process of an algorithm learning from the training data set can be thought of as a teacher supervising the learning process. The algorithm iteratively makes predictions on the training data and is corrected by the teacher. Learning stops when the algorithm achieves an acceptable level of performance.

Supervised Learning

- Supervised learning problems can be further grouped into regression and classification problems.
  - **Classification**: A classification problem is when the output variable is a category, such as “red” or “blue” or “disease” and “no disease”.
  - **Regression**: A regression problem is when the output variable is a real value, such as “dollars” or “weight”.

- One example is when you try to predict whether a tumour is malignant or benign according to the tumour size using a data set with tumour sizes and the type of tumour given to us.

Unsupervised Learning

- **Unsupervised learning** is where we only have input data (X) and no corresponding output data.

  The term unsupervised learning refers to the fact that the algorithm was provided with a data set in which we only specify the features of the instances while the class label information is not provided. The goal for unsupervised learning is to model the underlying structure or distribution in the data in order to learn more about the data. Algorithms are left to their own devises to discover and present the interesting structure in the data.

- One way to do it is to divide the given instances in the data sets into clusters. When a new instance is given the learning algorithm can predict which cluster this new instance belongs to. This is known as clustering algorithm.

Unsupervised Learning

- Unsupervised learning problems can be further grouped into clustering and association problems.
  - **Clustering**: A clustering problem is where you want to discover the inherent groupings in the data, such as grouping customers by purchasing behavior.
  - **Association**: An association rule learning problem is where we want to discover rules that describe large portions of your data, such as people that buy X also tend to buy Y.
- For example: Google news collects many new stories on the web and groups them together into different categories using this algorithm. This is done so that all stories about a single incident are grouped together.

Reinforcement Learning

- Reinforcement learning (RL) provide the learning agent with a reward function and let it figure out the best strategy for obtaining large rewards.
- **Reinforcement learning** is training by rewards and punishments. Here we train a computer as if we train a dog. If the dog obeys and acts according to our instructions we encourage it by giving biscuits or we punish it by beating or by scolding.
- Similarly, if the system works well then the teacher (environment) gives positive value (i.e. reward) or the teacher gives negative value (i.e. punishment). The learning system which gets the punishment has to improve itself.

Reinforcement Learning

- Thus it is a trial and error process.
- The *reinforcement learning algorithms* selectively retain the outputs that maximize the received reward over time.
- To accumulate a lot of rewards, the learning system must prefer the best experienced actions; however, it has to try new actions in order to discover better action selections for the future.
- Examples are learning to play games, robot control, elevator control, network routing, and animal learning.

Steps for Reinforcement Learning

- The agent observes an input state
- An action is determined by a decision making function (policy)
- The action is performed
- The agent receives a scalar reward or reinforcement from the environment
- Information about reward given for that state/action pair is recorded

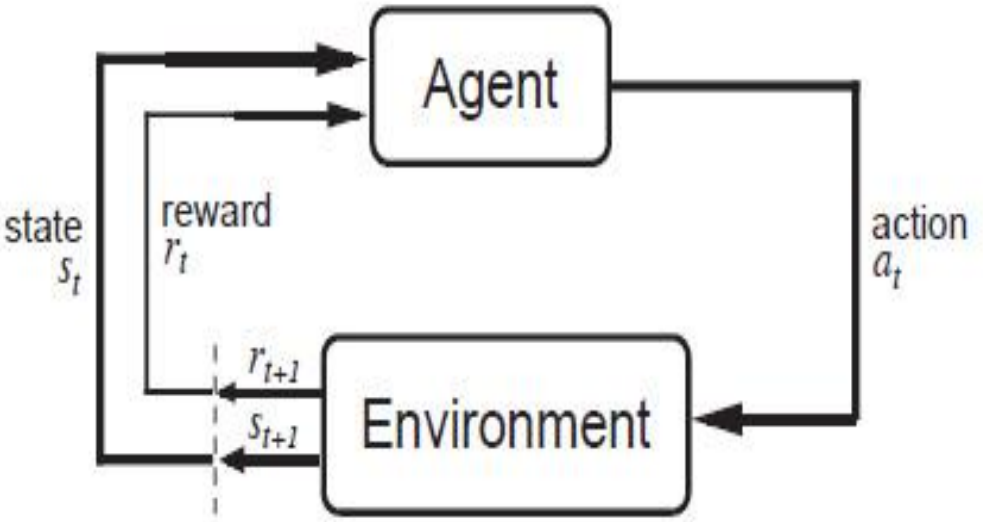


Figure: Agent Environment interface in Reinforcement Learning

Example of Reinforcement Learning

- **Personalization Shopping Support System** is one software that is designed to provide product information as per the user's interests.
  - It applies the reinforcement learning to analyze and learn customer behaviors and list out the products that the customers wish to buy.
  - If the system selects the right item that the customer wish to buy then it is given reward by assigning a particular value for the state and if the system selects an item which the user does not wish to buy then it is given the penalty.
  - This way the system learns the personal interests. In this process, the system acquires the knowledge of the user behavior and interest which makes it decide which information should be given to a particular user.
  - This results in greater customer satisfaction and increase in the success rate of product promotion.

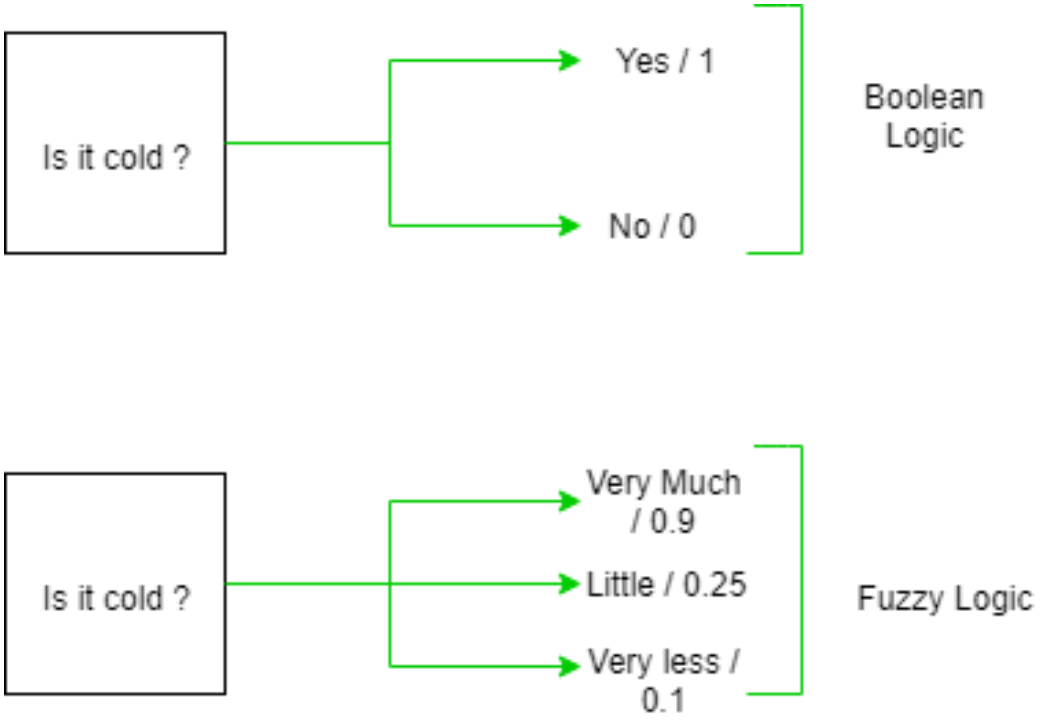
Fuzzy Learning

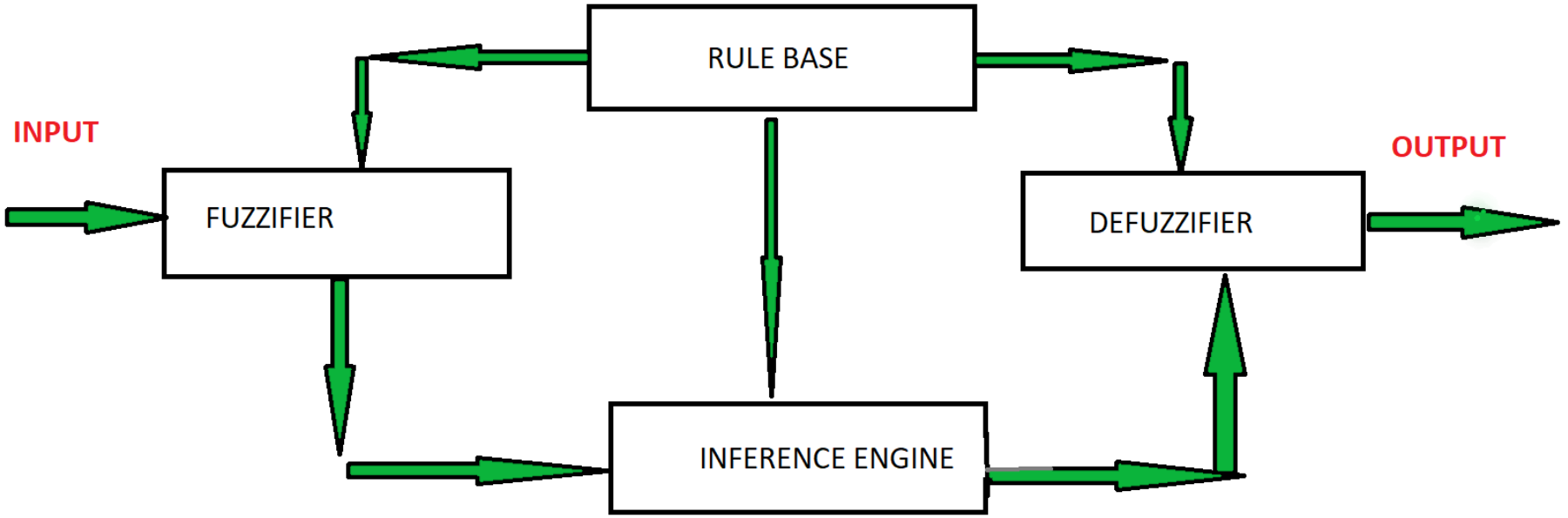
The term **fuzzy** refers to things which are not clear or are vague. In the real world many times we encounter a situation when we can't determine whether the state is true or false, their fuzzy logic provides a very valuable flexibility for reasoning.

In this way, we can consider the inaccuracies and uncertainties of any situation.

In Boolean system truth value, 1.0 represents absolute truth value and 0.0 represents absolute false value.

In the fuzzy system, there is no logic for absolute truth and absolute false value. But in fuzzy logic, there is intermediate value too present which is partially true and partially false.





FUZZY LOGIC ARCHITECTURE

RULE BASE:

It contains the set of rules and the IF-THEN conditions provided by the experts to govern the decision making system, on the basis of linguistic information.

FUZZIFICATION:

It is used to convert inputs i.e. crisp numbers into fuzzy sets. Crisp inputs are basically the exact inputs measured by sensors and passed into the control system for processing, such as temperature, pressure, rpm's, etc.

INFERENCE ENGINE:

It determines the matching degree of the current fuzzy input with respect to each rule and decides which rules are to be fired according to the input field.

DEFUZZIFICATION:

It is used to convert the fuzzy sets obtained by inference engine into a crisp value.

Application

- It is used in the aerospace field for altitude control of spacecraft and satellite.
- It has used in the automotive system for speed control, traffic control.
- It is used for decision making support systems and personal evaluation in the large company business.

LEARNING FROM EXAMPLES

- This involves the process of inductive learning-- where a system tries to induce a general rule from a set of observed instances.

- This involves classification -- assigning, to a particular input, the name of a class to which it belongs. Classification is important to many problem solving tasks.

- A learning system has to be capable of evolving its own class descriptions:

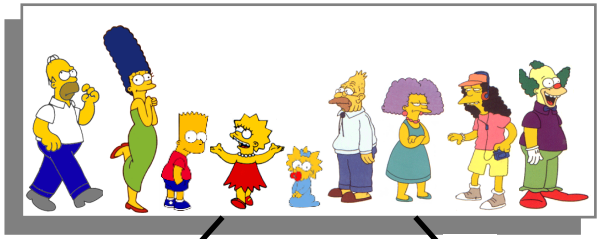
    - Initial class definitions may not be adequate.
    - The world may not be well understood or rapidly changing.

- The task of constructing class definitions is called induction or concept learning

- Example: ID3 Algorithm

Person	Hair Length	Weight	Age	Class
 Homer	0”	250	36	**M**
 Marge	10”	150	34	**F**
 Bart	2”	90	10	**M**
 Lisa	6”	78	8	**F**
 Maggie	4”	20	1	**F**
 Abe	1”	170	70	**M**
 Selma	8”	160	41	**F**
 Otto	10”	180	38	**M**
 Krusty	6”	200	45	**M**

 Comic	8”	290	38	**?**



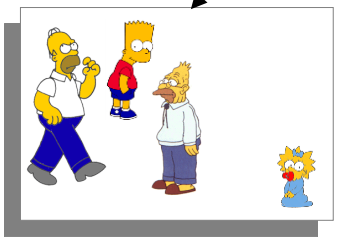
$$Entropy(S) = -\frac{p}{p+n}\log_2\left(\frac{p}{p+n}\right) - \frac{n}{p+n}\log_2\left(\frac{n}{p+n}\right)$$

$Entropy(4\mathbf{F},5\mathbf{M}) = -(4/9)\log_2(4/9) - (5/9)\log_2(5/9)$
$= \mathbf{0.9911}$

yes

no

Hair Length <= 5?





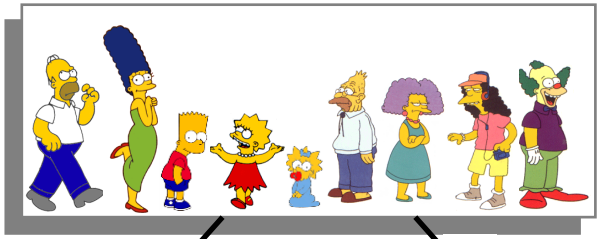
Let us try splitting on *Hair length*

$Entropy(1\mathbf{F},3\mathbf{M}) = -(1/4)\log_2(1/4) - (3/4)\log_2(3/4)$
$= \mathbf{0.8113}$

$Entropy(3\mathbf{F},2\mathbf{M}) = -(3/5)\log_2(3/5) - (2/5)\log_2(2/5)$
$= \mathbf{0.9710}$

$$Gain(A) = E(Current\ set) - \sum E(all\ child\ sets)$$

$Gain(\text{Hair Length} <= 5) = \mathbf{0.9911} - (4/9 * \mathbf{0.8113} + 5/9 * \mathbf{0.9710}) = \mathbf{0.0911}$



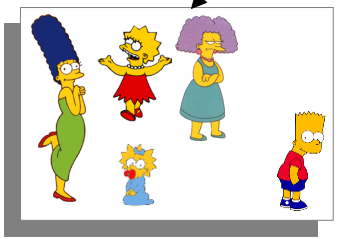
$$Entropy(S) = -\frac{p}{p+n}\log_2\left(\frac{p}{p+n}\right) - \frac{n}{p+n}\log_2\left(\frac{n}{p+n}\right)$$

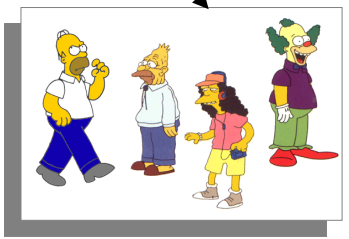
$Entropy(4\mathbf{F},5\mathbf{M}) = -(4/9)\log_2(4/9) - (5/9)\log_2(5/9)$
$= \mathbf{0.9911}$

yes

no

Weight <= 160?





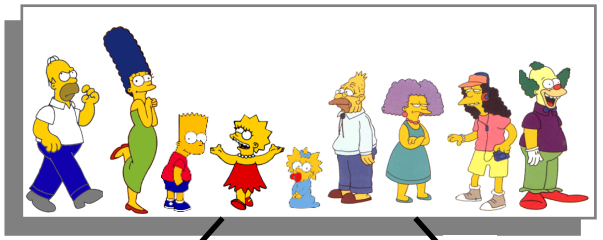
Let us try splitting on *Weight*

$Entropy(4\mathbf{F},1\mathbf{M}) = -(4/5)\log_2(4/5) - (1/5)\log_2(1/5)$
$= \mathbf{0.7219}$

$Entropy(0\mathbf{F},4\mathbf{M}) = -(0/4)\log_2(0/4) - (4/4)\log_2(4/4)$
$= \mathbf{0}$

$$Gain(A) = E(Current\ set) - \sum E(all\ child\ sets)$$

$Gain(\text{Weight <= 160}) = \mathbf{0.9911} - (5/9 * \mathbf{0.7219} + 4/9 * \mathbf{0}) = \mathbf{0.5900}$



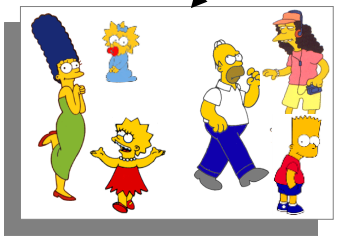
$$Entropy(S) = -\frac{p}{p+n}\log_2\left(\frac{p}{p+n}\right) - \frac{n}{p+n}\log_2\left(\frac{n}{p+n}\right)$$

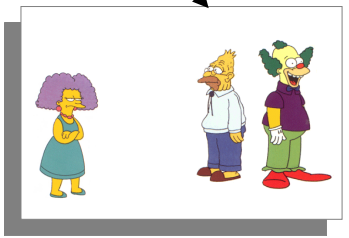
$Entropy(4\mathbf{F},5\mathbf{M}) = -(4/9)\log_2(4/9) - (5/9)\log_2(5/9)$
$= \mathbf{0.9911}$

yes

no

age <= 40?





Let us try splitting on *Age*

$Entropy(3\mathbf{F},3\mathbf{M}) = -(3/6)\log_2(3/6) - (3/6)\log_2(3/6)$
$= \mathbf{1}$

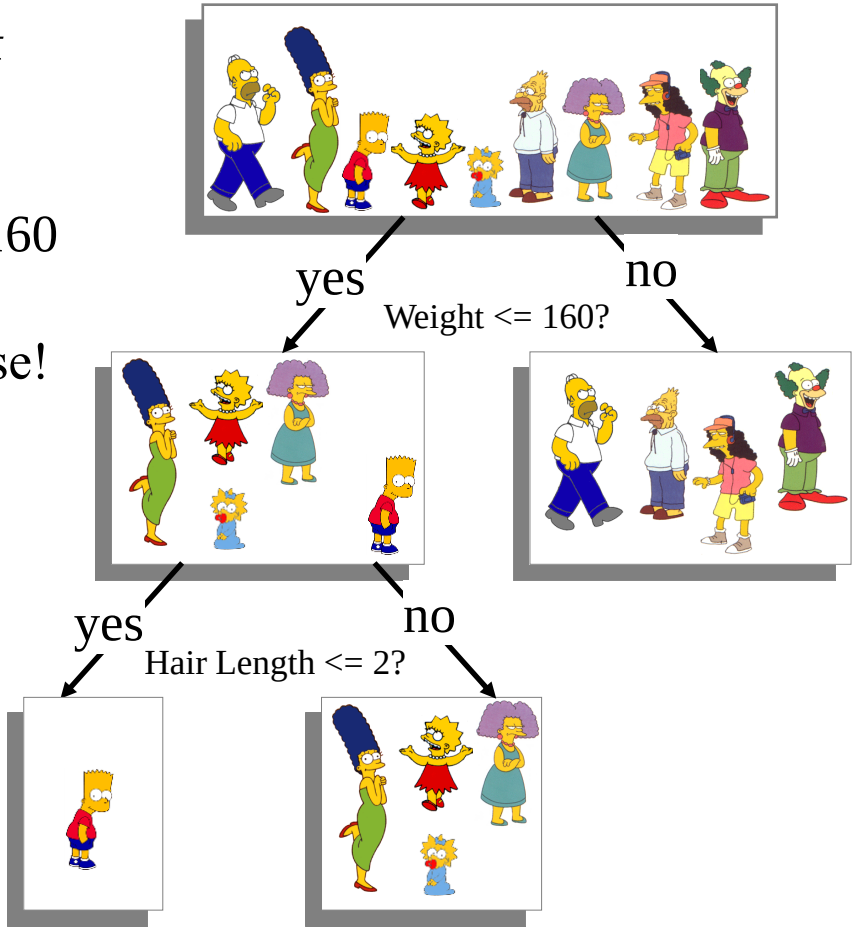
$Entropy(1\mathbf{F},2\mathbf{M}) = -(1/3)\log_2(1/3) - (2/3)\log_2(2/3)$
$= \mathbf{0.9183}$

$$Gain(A) = E(Current\ set) - \sum E(all\ child\ sets)$$

$Gain(\text{Age} <= 40) = \mathbf{0.9911} - (6/9 * \mathbf{1} + 3/9 * \mathbf{0.9183}) = \mathbf{0.0183}$

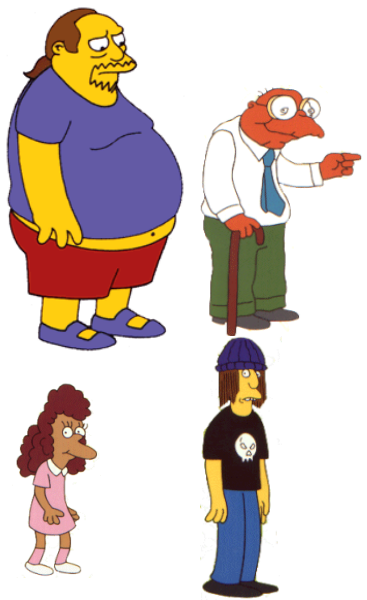
Of the 3 features we had, *Weight* was best. But while people who weigh over 160 are perfectly classified (as males), the under 160 people are not perfectly classified… So we simply recurse!

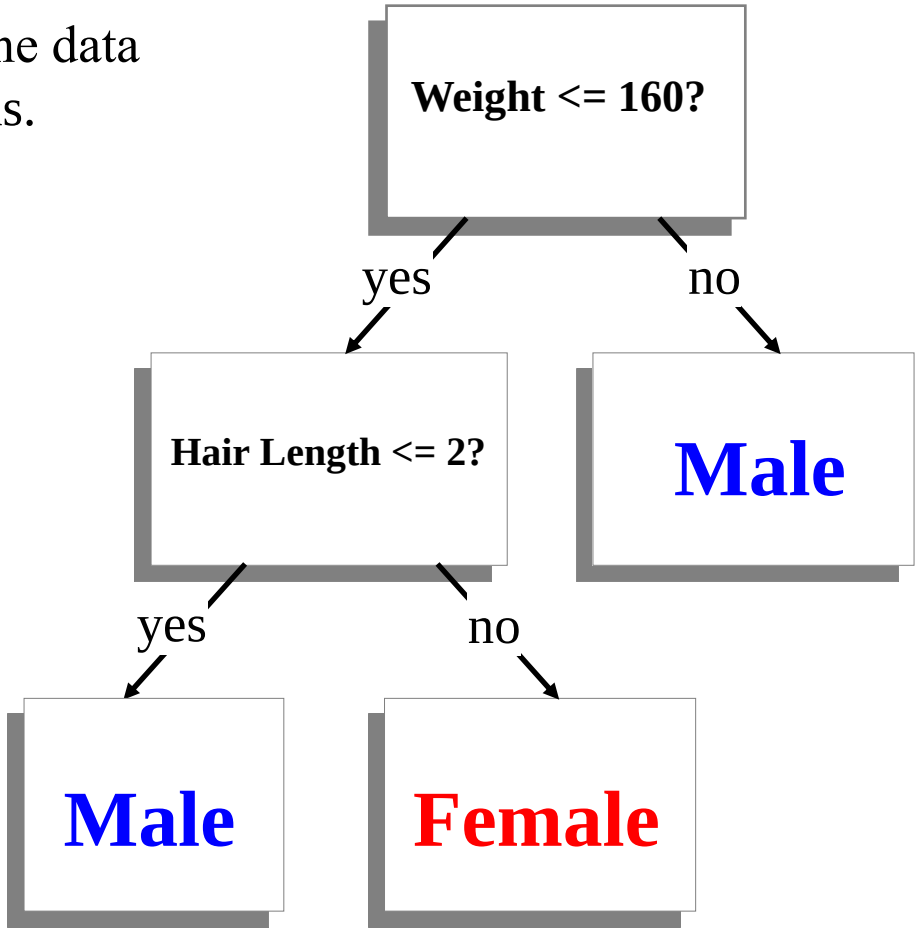
This time we find that we can split on *Hair length*, and we are done!



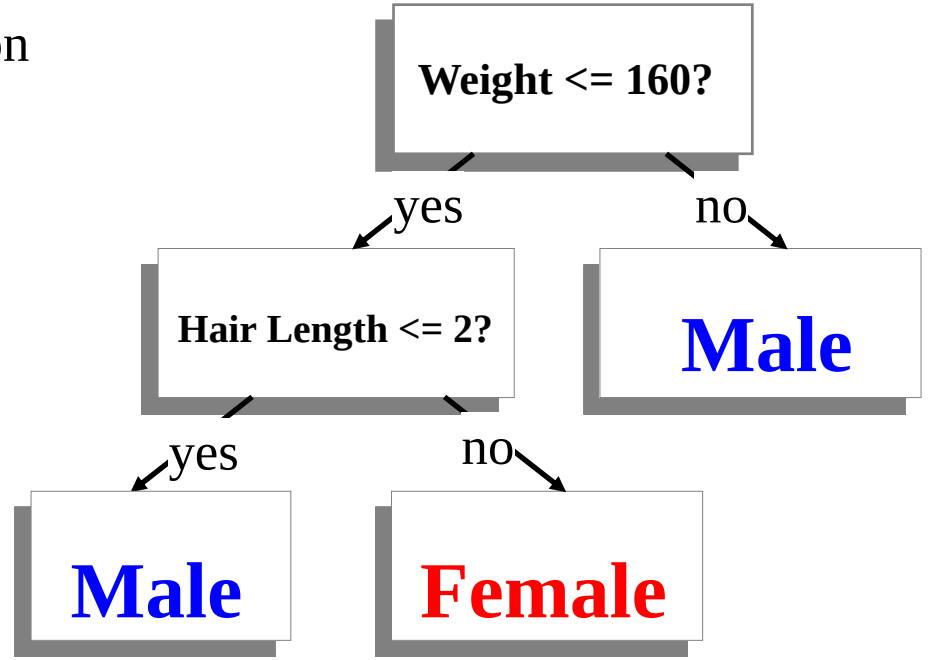
We need don't need to keep the data around, just the test conditions.

How would these people be classified?





It is trivial to convert Decision Trees to rules…

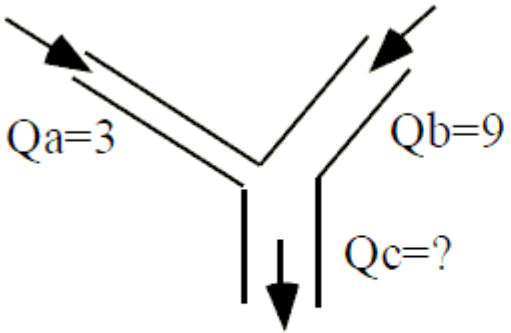


Rules to Classify Males/Females

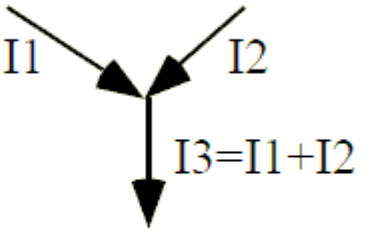
If *Weight* **greater than** 160, classify as **Male**
Elseif *Hair Length* **less than or equal** to 2, classify as **Male**
Else classify as **Female**

Learning by analogy

Learning by analogy means acquiring new knowledge about an input entity by transferring it from a known similar entity



Simple Hydraulics Problem



Kirchoff's First Law

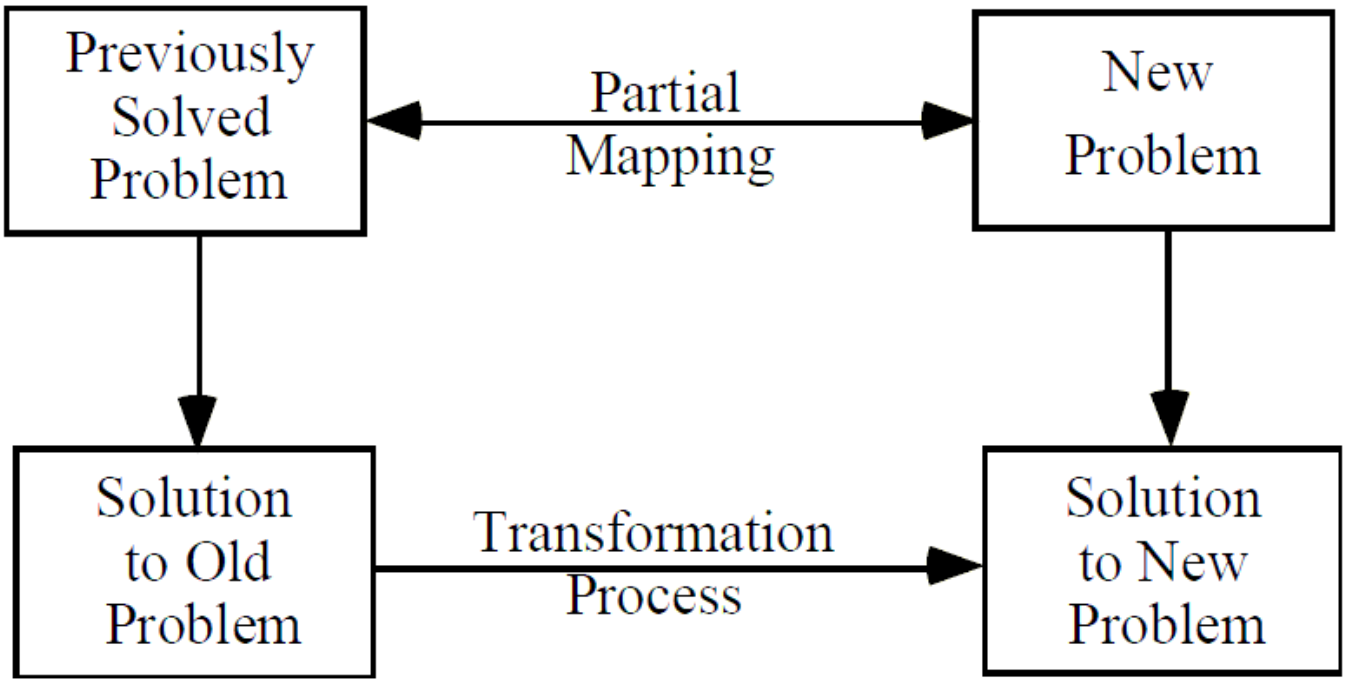
One may infer, by analogy, that hydraulics laws are similar to Kirchhoff's laws

Problem solving by analogy

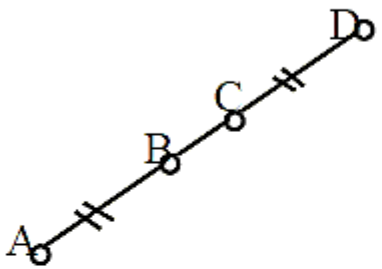
Analogy means deriving new knowledge about an input entity by transferring it from a known similar entity.

How could we define problem solving by analogy?

Problem solving by analogy is the process of transferring knowledge from past problem-solving episodes to new problems that share significant aspects with corresponding past experience and using the transferred knowledge to construct solutions to the new problems.

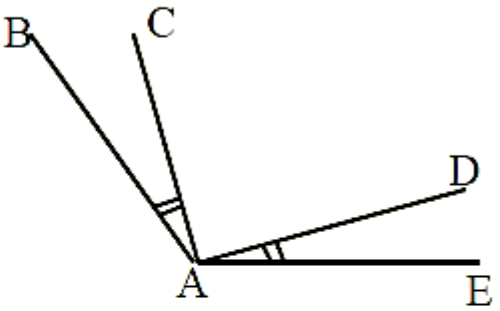


GIVEN: AB = CD
PROVE: AC = BD



σ = (AB <- <BAC
CD <- <DAE
AC <- <BAD
BD <- <CAE)

GIVEN: <BAC = <DAE
PROVE: <BAD = <CAE



AB = CD
BC = BC
AB + BC = BC + CD
AC = BD

<BAC = <DAE
<CAD = <CAD
<BAC + <CAD= <CAD + <DAE
<BAD = <CAE

Learning By Simulated Evolution

- Evolving a solution
- Begin with population of individuals
  - –Individuals = candidate solutions ~chromosomes
- Produce offspring with variation
  - –Mutation: change features
  - –Crossover: exchange features between individuals
- Apply natural selection
  - –Select “best” individuals to go on to next generation Continue until satisfied with solution

Genetic Algorithm

Genetic Algorithm (GA) is a search-based optimization technique based on the principles of Genetics and Natural Selection.

It is frequently used to solve optimization problems, in research, and in machine learning.

Optimization is the process of making something better.

Optimization refers to finding the values of inputs in such a way that we get the “best” output values. The definition of “best” varies from problem to problem, but in mathematical terms, it refers to maximizing or minimizing one or more objective functions, by varying the input parameters.

A genetic algorithm is a search heuristic that is inspired by Charles Darwin’s theory of natural evolution.

This algorithm reflects the process of natural selection where the fittest individuals are selected for reproduction in order to produce offspring of the next generation.

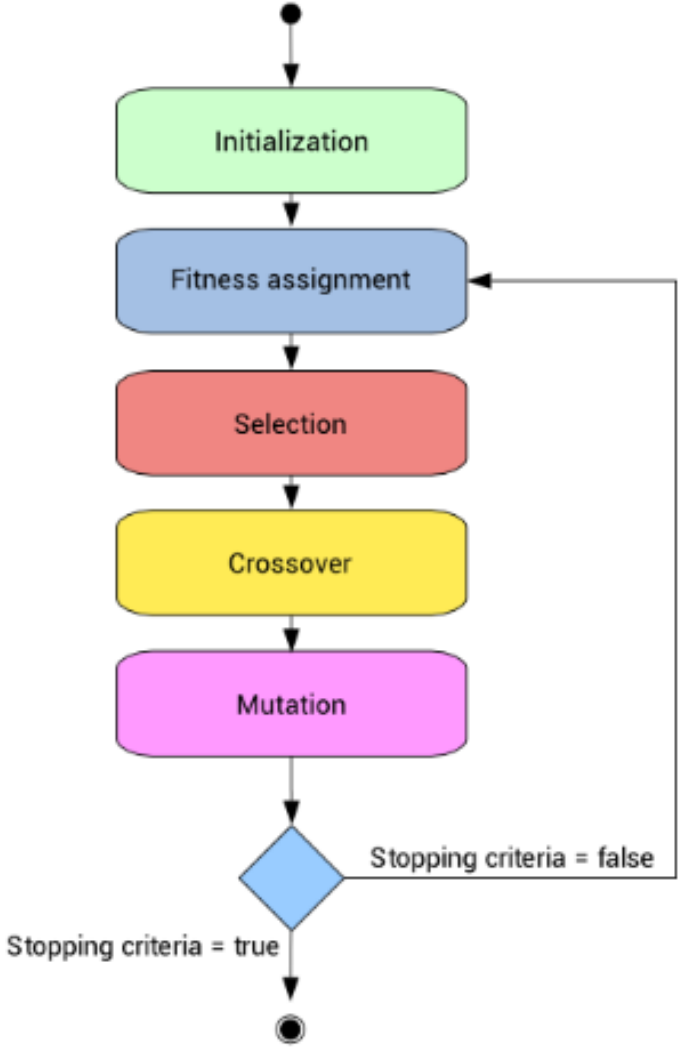
The process of natural selection starts with the selection of fittest individuals from a population.

They produce offspring which inherit the characteristics of the parents and will be added to the next generation. If parents have better fitness, their offspring will be better than parents and have a better chance at surviving. This process keeps on iterating and at the end, a generation with the fittest individuals will be found.

This notion can be applied for a search problem. We consider a set of solutions for a problem and select the set of best ones out of them.

Five phases are considered in a genetic algorithm.

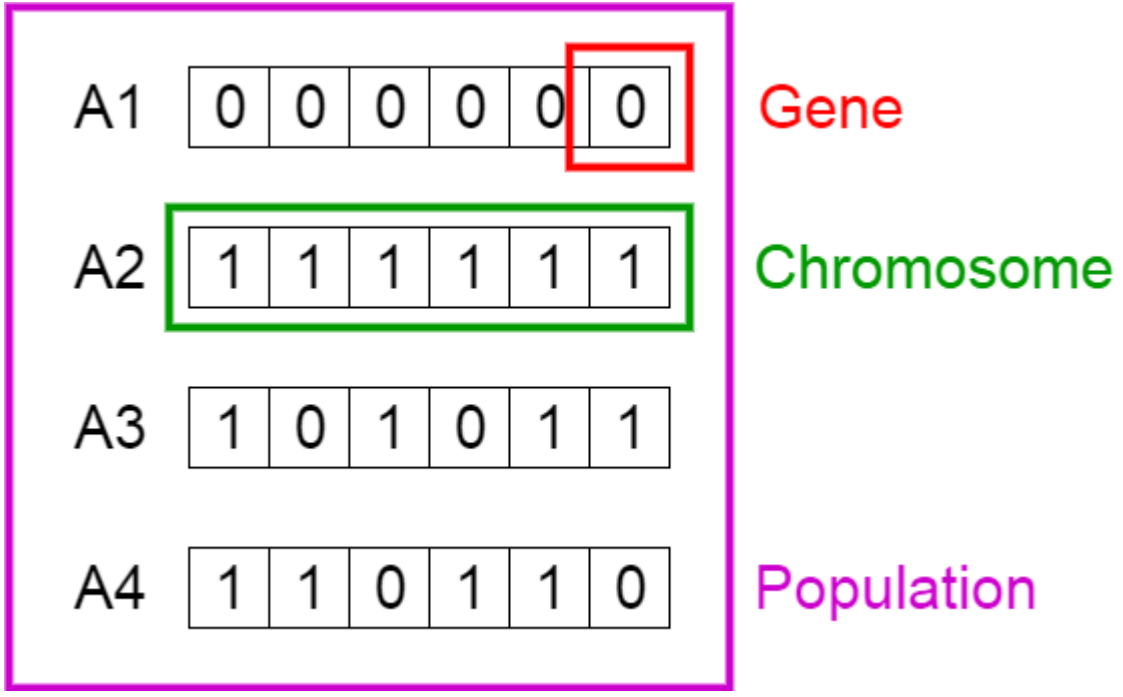
1. Initial population
2. Fitness function
3. Selection
4. Crossover
5. Mutation


Initialization
Fitness assignment
Selection
Crossover
Mutation
Stopping criteria = false
Stopping criteria = true

Initial Population

The process begins with a set of individuals which is called a Population. Each individual is a solution to the problem we want to solve.

An individual is characterized by a set of parameters (variables) known as Genes. Genes are joined into a string to form a Chromosome (solution).



Fitness Function

The fitness function determines how fit an individual is (the ability of an individual to compete with other individuals). It gives a fitness score to each individual. The probability that an individual will be selected for reproduction is based on its fitness score.

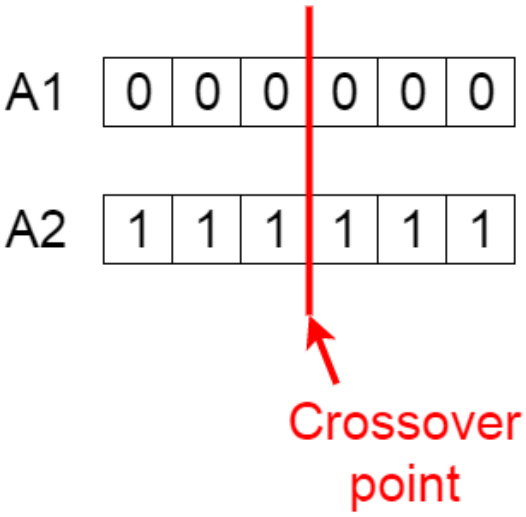
Selection

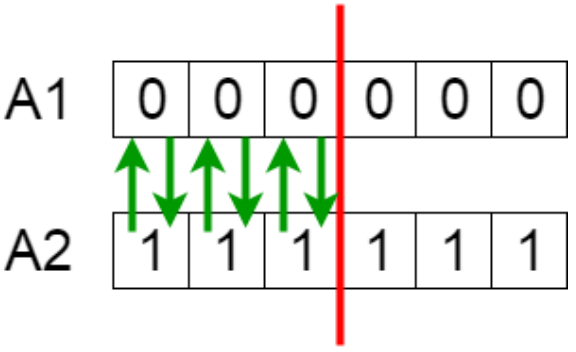
The idea of selection phase is to select the fittest individuals and let them pass their genes to the next generation.

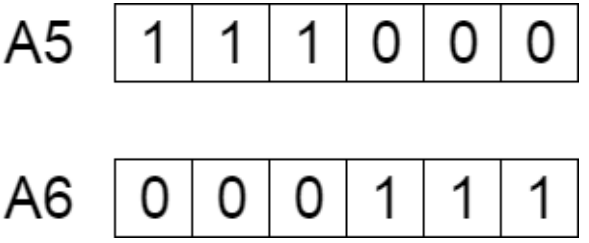
Two pairs of individuals (parents) are selected based on their fitness scores. Individuals with high fitness have more chance to be selected for reproduction.

Crossover

Crossover is the most significant phase in a genetic algorithm. For each pair of parents to be mated, a crossover point is chosen at random from within the genes.

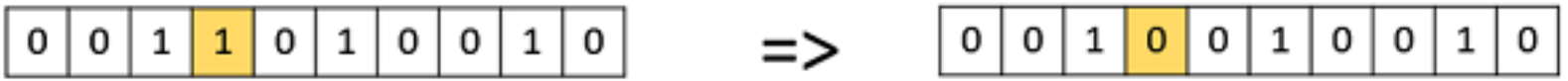






Mutation

In certain new offspring formed, some of their genes can be subjected to a mutation with a low random probability. This implies that some of the bits in the bit string can be flipped.



Termination

The algorithm terminates if the population has converged (does not produce offspring which are significantly different from the previous generation).

Then it is said that the genetic algorithm has provided a set of solutions to our problem.

Assignment

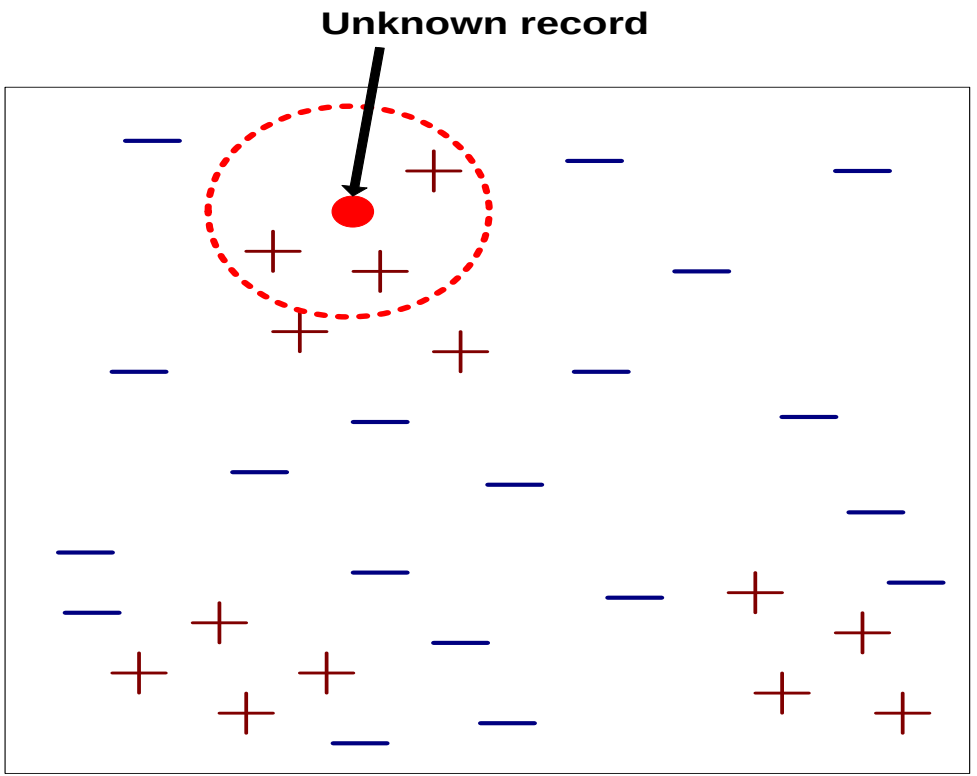
- ❑ Explain the process how the neural Network Learns
- ❑ Write Short Notes on Learning by Training Perceptron

If you want to READ extra

K-Nearest Neighbor (KNN)

- KNN is a simple algorithm that stores all available cases and classifies new cases based on a similarity measure (usually Euclidean Distance)
- It can be used both for classification and regression
- KNN is a "**non parametric**" "**lazy learning**" algorithm
  - Non parametric means that it does not make any assumptions on the underlying data distribution
  - Lazy algorithm because there is zero effort at training time and all the effort at prediction time.

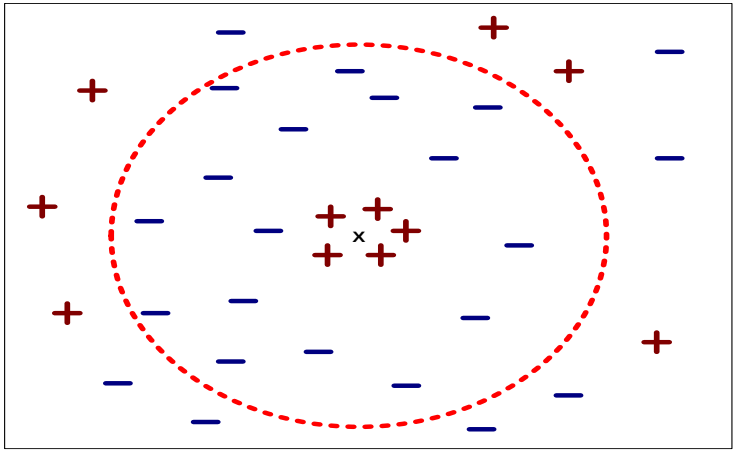
K-Nearest Neighbor Classifiers

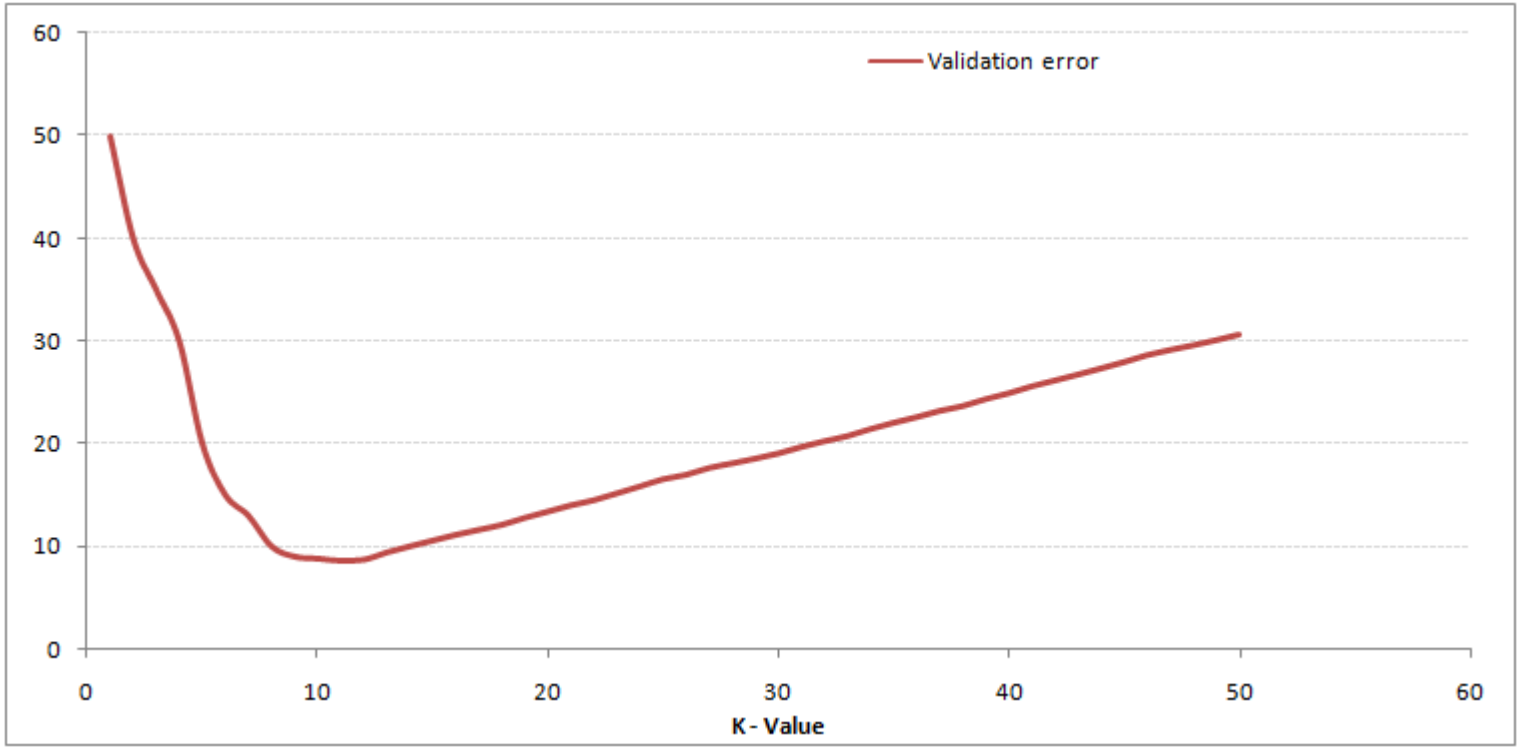


- Requires **three** things
  - The set of stored records
  - Distance Metric to compute distance between records
  - The value of *k*, the number of nearest neighbors to retrieve
- To classify an unknown record:
  - Compute distance (Euclidean distance) to other training records
  - Identify *k* nearest neighbors
  - Use class labels of nearest neighbors to determine the class label of unknown record (e.g., by taking majority vote)

K-Nearest Neighbor Classification

- Choosing the value of k:
  - If k is too small, sensitive to noise points
  - If k is too large, neighborhood may include points from other classes





To get the optimal value of K, we can separate the training and validation from the initial data set and plot the validation error curve to get the optimal value of K

Suppose we have height, weight and T-shirt size of some customers.

We need to predict the T-shirt size of a *new customer* given only height (161cm) and weight(61kg)

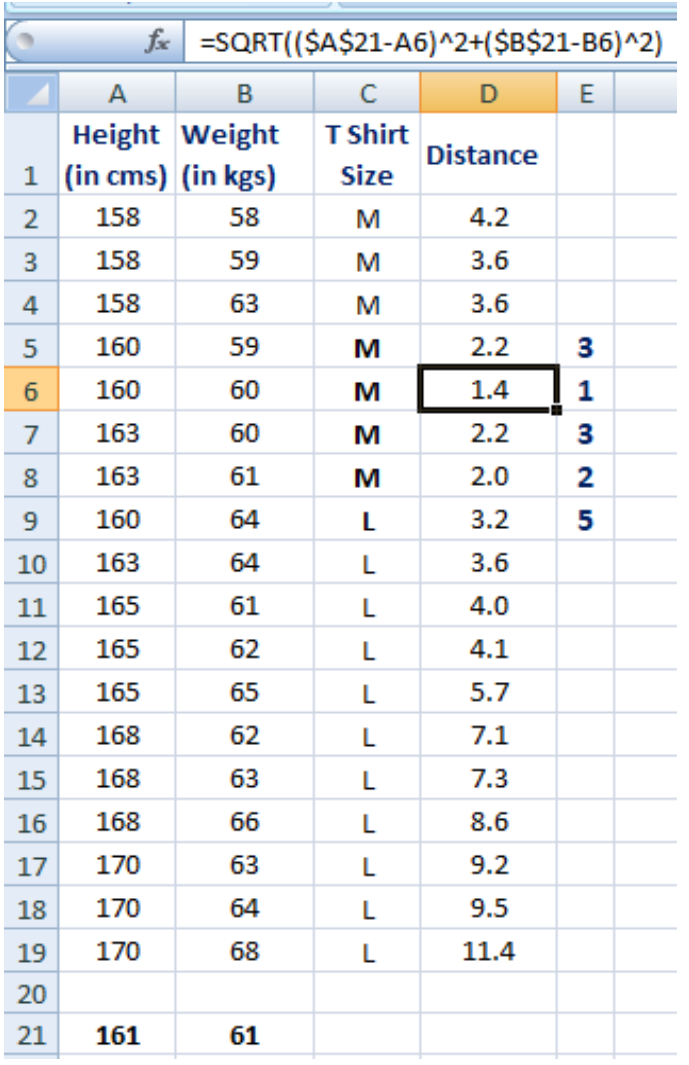
Euclidean :

$$d(x, y) = \sqrt{\sum_{i=1}^{m} (x_i - y_i)^2}$$

Height (in cms)	Weight (in kgs)	T Shirt Size
158	58	M
158	59	M
158	63	M
160	59	M
160	60	M
163	60	M
163	61	M
160	64	L
163	64	L
165	61	L
165	62	L
165	65	L
168	62	L
168	63	L
168	66	L
170	63	L
170	64	L
170	68	L

Let k be 5.

Then the algorithm searches for the 5 customers closest to *new customer*, i.e. most similar to new customer in terms of attributes, and see what categories those 5 customers were in. If 4 of them had ‘Medium T shirt sizes’ and 1 had ‘Large T shirt size’ then your best guess for new customer is ‘Medium T shirt.



	A	B	C	D	E
1	Height (in cms)	Weight (in kgs)	T Shirt Size	Distance	
2	158	58	M	4.2	
3	158	59	M	3.6	
4	158	63	M	3.6	
5	160	59	**M**	2.2	**3**
6	160	60	**M**	1.4	**1**
7	163	60	**M**	2.2	**3**
8	163	61	**M**	2.0	**2**
9	160	64	**L**	3.2	**5**
10	163	64	L	3.6	
11	165	61	L	4.0	
12	165	62	L	4.1	
13	165	65	L	5.7	
14	168	62	L	7.1	
15	168	63	L	7.3	
16	168	66	L	8.6	
17	170	63	L	9.2	
18	170	64	L	9.5	
19	170	68	L	11.4	
20					
21	**161**	**61**			

Naive Bayes

- Naive Bayes is a probabilistic machine learning algorithm based on the Bayes Theorem, used in a wide variety of classification tasks.
- Typical applications include filtering spam, classifying documents, sentiment prediction etc.
- Naive (showing a lack of experience, wisdom, or judgement) is named so because it assumes the features that go into the model is independent of each other. That is changing the value of one feature, does not directly influence or change the value of any of the other features used in the algorithm.

There are two parts to this algorithm:

- **Naïve**
  independence
- **Bayes**
  Bayesian Theorem

What is Bayes Theorem?

In Statistics and probability theory, Bayes' theorem describes the probability of an event, based on prior knowledge of conditions that might be related to the event. It serves as a way to figure out conditional probability.

Given a Hypothesis **H** and evidence **E,** Bayes' Theorem states that the relationship between the probability of Hypothesis before getting the evidence **P(H)** and the probability of the hypothesis after getting the evidence **P(H|E)** is :

$$P(H|E) = \frac{P(E|H).P(H)}{P(E)}$$

This relates the probability of the hypothesis before getting the evidence **P(H)**, to the probability of the hypothesis after getting the evidence, **P(H|E)**. For this reason, is called the **prior probability**, while **P(H|E)** is called the **posterior probability**. The factor that relates the two, **P(H|E) / P(E)**, is called the **likelihood ratio**. Using these terms, Bayes' theorem can be rephrased as:

"The posterior probability equals the prior probability times the likelihood ratio."

Numerical

See Chapter 4

Game Prediction using Bayes' Theorem

we have our Data, which comprises of the Day, Outlook, Humidity, Wind Conditions and the final column being Play, which we have to predict.

Day	Outlook	Humidity	Wind	Play
D1	Sunny	High	Weak	No
D2	Sunny	High	Strong	No
D3	Overcast	High	Weak	Yes
D4	Rain	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	Strong	No
D7	Overcast	Normal	Strong	Yes
D8	Sunny	High	Weak	No
D9	Sunny	Normal	Weak	Yes
D10	Rain	Normal	Weak	Yes
D11	Sunny	Normal	Strong	Yes
D12	Overcast	High	Strong	Yes
D13	Overcast	Normal	Weak	Yes
D14	Rain	High	Strong	No

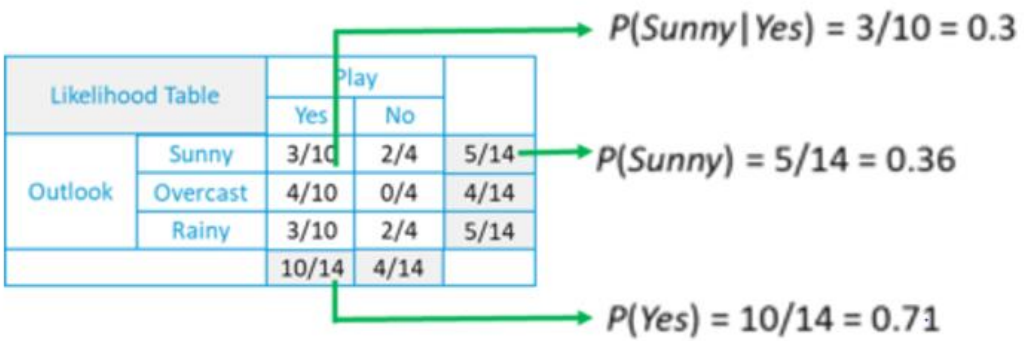
First, we will create a **frequency** table using each attribute of the dataset.

Frequency Table		Play	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	3	2

Frequency Table		Play	
		Yes	No
Humidity	High	3	4
	Normal	6	1

Frequency Table		Play	
		Yes	No
Wind	Strong	6	2
	Weak	3	3

For each frequency table, we will generate a **likelihood** table.



Likelihood Table		Play		
		Yes	No	
Outlook	Sunny	3/10	2/4	5/14
	Overcast	4/10	0/4	4/14
	Rainy	3/10	2/4	5/14
		10/14	4/14	

Likelihood of '**Yes**' given '**Sunny**' is:

P(Yes|Sunny) = P(Sunny|Yes)* P(Yes) / P(Sunny) = (0.3 x 0.71) /0.36 = 0.591

Similarly Likelihood of '**No**' given '**Sunny**' is:

P(No|Sunny) = P(Sunny|No)* P(No) / P(Sunny) = (0.4 x 0.36) /0.36 = 0.40

Likelihood table for Humidity

Likelihood Table		Play		
		Yes	No	
Humidity	High	3/9	4/5	7/14
	Normal	6/9	1/5	7/14
		9/14	5/14	

$P(Yes|High) = 0.33 \times 0.6 / 0.5 = 0.42$

$P(No|High) = 0.8 \times 0.36 / 0.5 = 0.58$

Likelihood table for Wind

Likelihood Table		Play		
		Yes	No	
Wind	Weak	6/9	2/5	8/14
	Strong	3/9	3/5	6/14
		9/14	5/14	

$P(Yes|Weak) = 0.67 \times 0.64 / 0.57 = 0.75$

$P(No|Weak) = 0.4 \times 0.36 / 0.57 = 0.25$

Suppose we have a **Day** with the following values :

- **Outlook = Rain**
- **Humidity = High**
- **Wind = Weak**
- **Play =?**

So, with the data, we have to predict whether “we can play on that day or not”.

Likelihood of ‘Yes’ on that Day = P(Outlook = Rain|Yes)*P(Humidity= High|Yes)* P(Wind= Weak|Yes)*P(Yes)

= **2/9 * 3/9 * 6/9 * 9/14 = 0.0199**

Likelihood of ‘No’ on that Day = P(Outlook = Rain|No)*P(Humidity= High|No)* P(Wind= Weak|No)*P(No)

= **2/5 * 4/5 * 2/5 * 5/14 = 0.0166**

Now we normalize the values, then

P(Yes) = 0.0199 / (0.0199+ 0.0166) = 0.55

P(No) = 0.0166 / (0.0199+ 0.0166) = 0.45

Our model predicts that there is a **55%** chance there will be a Game tomorrow.

Naive Bayes in the Industry

- News Categorization
- Spam Filtering
- Weather Forecasting
- Medical Diagosis

Logistic regression

(Regression is a measure of the relation between the mean value of one variable (e.g. output) and corresponding values of other variables)

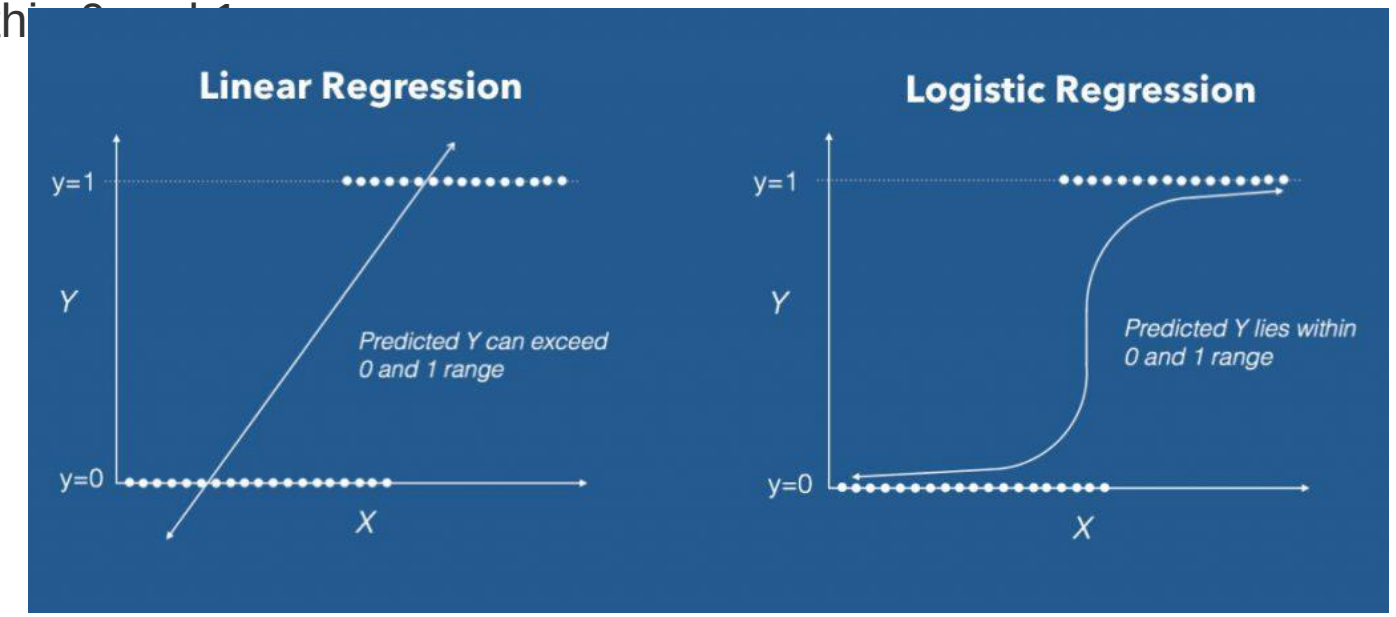
Logistic regression is a predictive modelling algorithm that is used when the Y variable is binary categorical. That is, it can take only two values like 1 or 0.

The goal is to determine a mathematical equation that can be used to predict the probability of event 1. Once the equation is established, it can be used to predict the Y when only the X's are known.

Why not linear regression?

When the response variable has only 2 possible values, it is desirable to have a model that predicts the value either as 0 or 1 or as a probability score that ranges between 0 and 1.

Linear regression does *not* have this capability. Because, If we use linear regression to model a binary response variable, the resulting model may not restrict the predicted Y values with



Earlier we saw what is linear regression and how to use it to predict continuous Y variables.
In linear regression the Y variable is always a continuous variable. If suppose, the Y variable was categorical, you cannot use linear regression model it.

So what would you do when the Y is a categorical variable with 2 classes?

Logistic regression can be used to model and solve such problems, also called as binary classification problems.

A key point to note here is that Y can have 2 classes only and not more than that. If Y has more than 2 classes, it would become a multi class classification and you can no longer use the vanilla logistic regression for that.

Yet, Logistic regression is a classic predictive modelling technique and still remains a popular choice for modelling binary categorical variables.
Another advantage of logistic regression is that it computes a prediction probability score of an event. More on that when you actually start building the models.

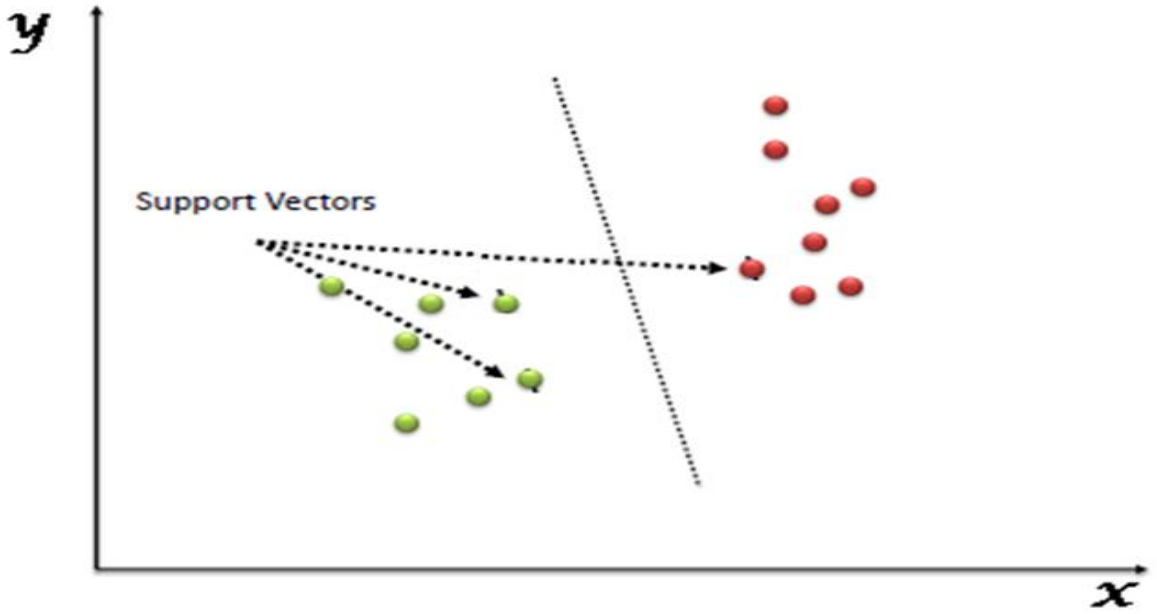
Here are some examples of binary classification problems:

- **Spam Detection** : Predicting if an email is Spam or not
- **Credit Card Fraud** : Predicting if a given credit card transaction is fraud or not
- **Health** : Predicting if a given mass of tissue is benign or malignant
- **Marketing** : Predicting if a given user will buy an insurance product or not

Support Vector Machines (SVM)

“Support Vector Machine” (SVM) is a supervised machine learning algorithm which can be used for both classification or regression challenges. However, it is mostly used in classification problems.

In this algorithm, we plot each data item as a point in n-dimensional space (where n is number of features you have) with the value of each feature being the value of a particular coordinate. Then, we perform classification by finding the hyper-plane that differentiate the two classes very well

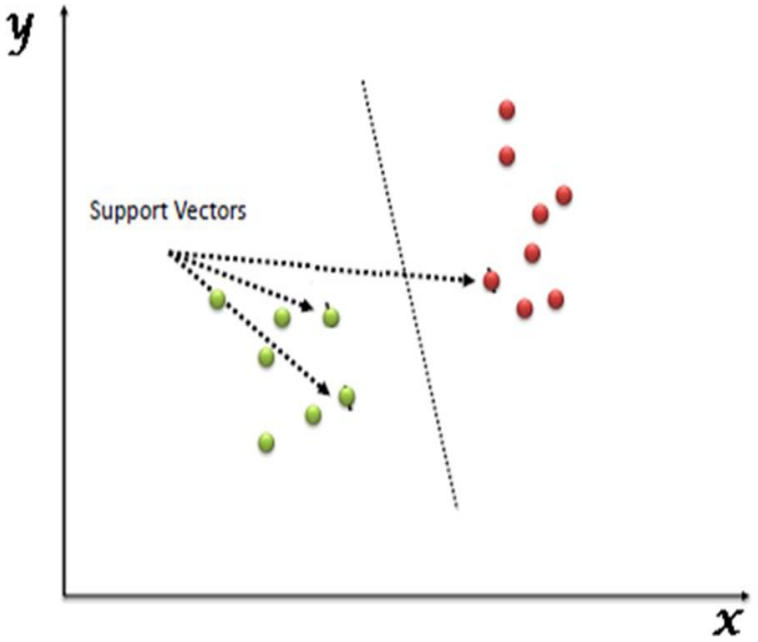


Support Vectors

Support vectors are the data points nearest to the hyperplane, the points of a data set that, if removed, would alter the position of the dividing hyperplane. Because of this, they can be considered the critical elements of a data set.

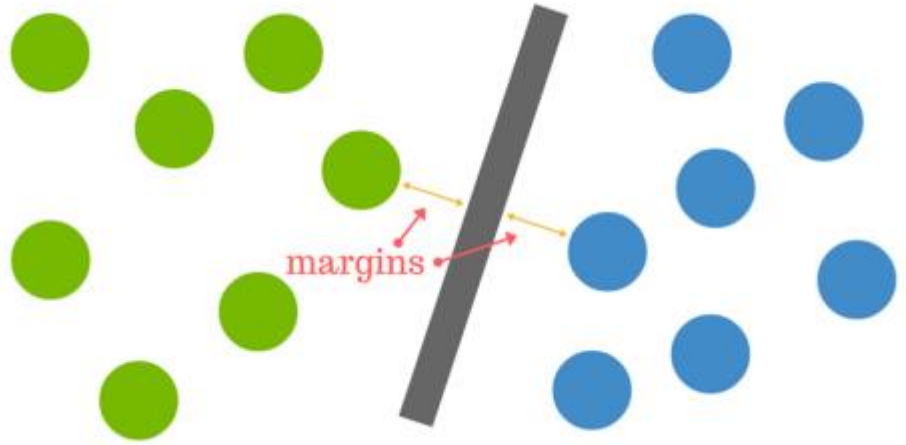
What is a hyperplane?

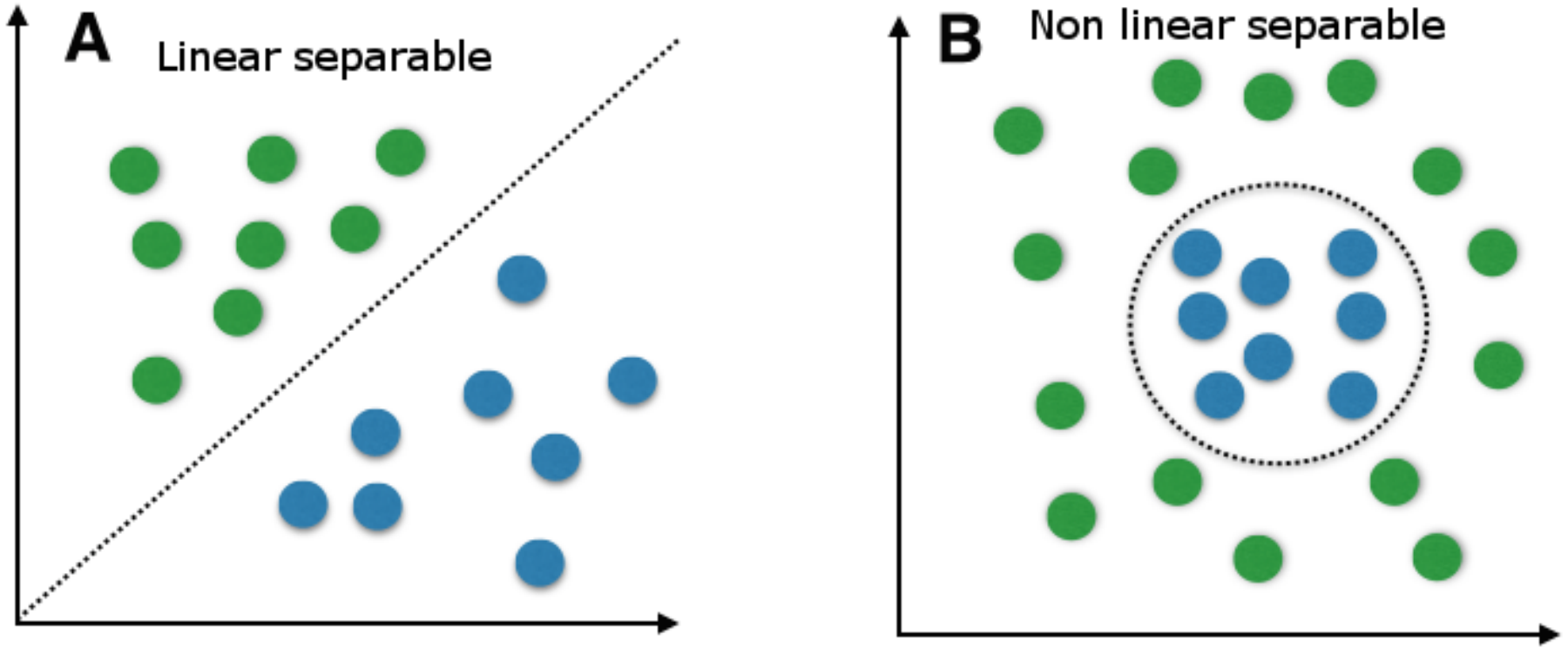
As a simple example, for a classification task with only two features (like the image), we can think of a hyperplane as a line that linearly separates and classifies a set of data.



How do we find the right hyperplane?

- Or, in other words, how do we best segregate the two classes within the data?
- The distance between the hyperplane and the nearest data point from either set is known as the margin. The goal is to choose a hyperplane with the greatest possible margin between the hyperplane and any point within the training set, giving a greater chance of new data being classified correctly.

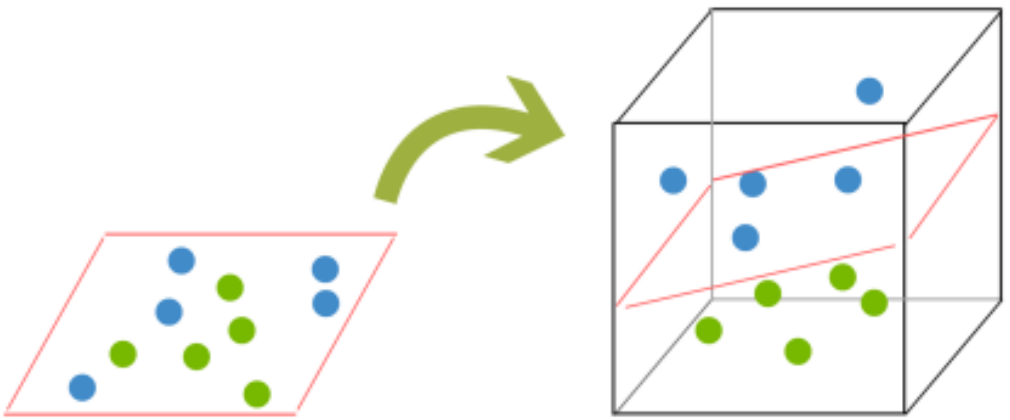



A
Linear separable
B
Non linear separable

But what happens when there is no clear hyperplane?



In order to classify a dataset like the one above it’s necessary to move away from a 2d view of the data to a 3d view. Explaining this is easiest with another simplified example. Imagine that our two sets of colored balls above are sitting on a sheet and this sheet is lifted suddenly, launching the balls into the air. While the balls are up in the air, you use the sheet to separate them. This ‘lifting’ of the balls represents the mapping of data into a higher dimension. This is known as kernelling



Because we are now in three dimensions, our hyperplane can no longer be a line. It must now be a plane as shown in the example above. The idea is that the data will continue to be mapped into higher and higher dimensions until a hyperplane can be formed to segregate it.

Pros

- ❑ Accuracy
- ❑ Works well on smaller cleaner datasets
- ❑ It can be more efficient because it uses a subset of training points

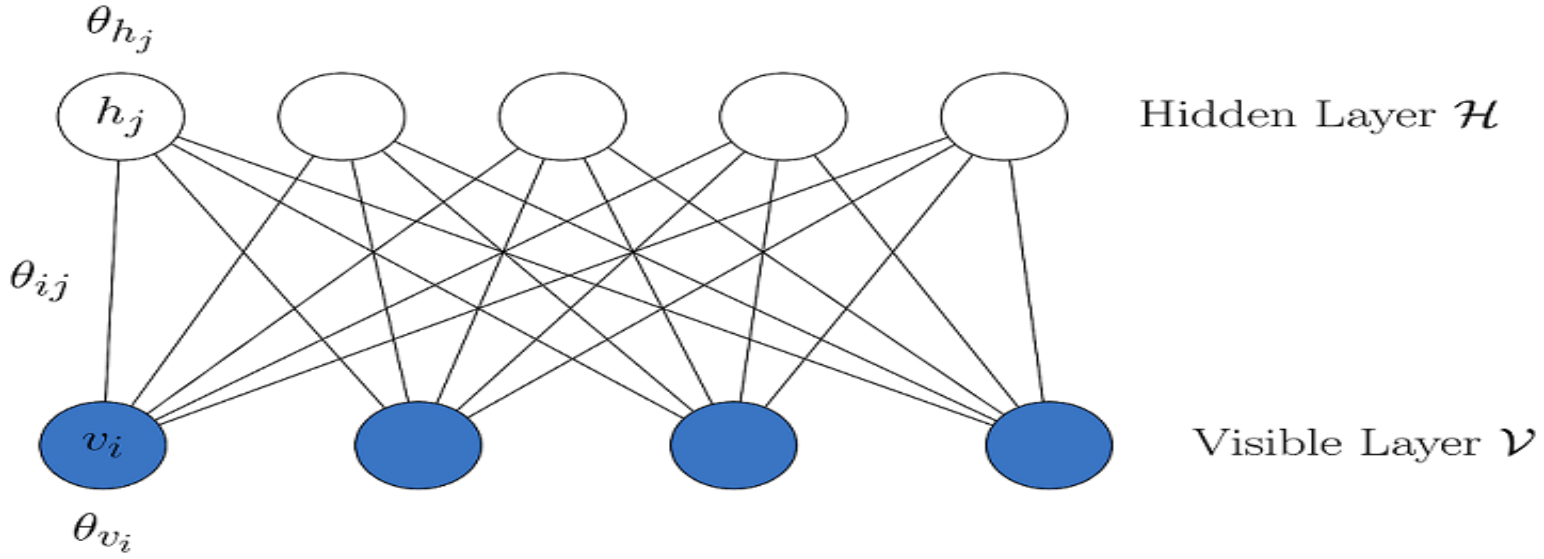
Cons

- ❑ Isn’t suited to larger datasets as the training time with SVMs can be high
- ❑ Less effective on noisier datasets with overlapping classes

Boltzmann Machines

- A Hopfield Net consisting of Binary Stochastic Neuron with hidden units is called Boltzmann Machine.
- A Boltzmann Machine is a network of symmetrically connected, neuron like units that make stochastic decisions about whether to be on or off.

Structure of Boltzmann Machine



- The stochastic neurons of Boltzmann machine are in two groups: visible and hidden.
- Visible neurons provide an interface between the net and its environment.

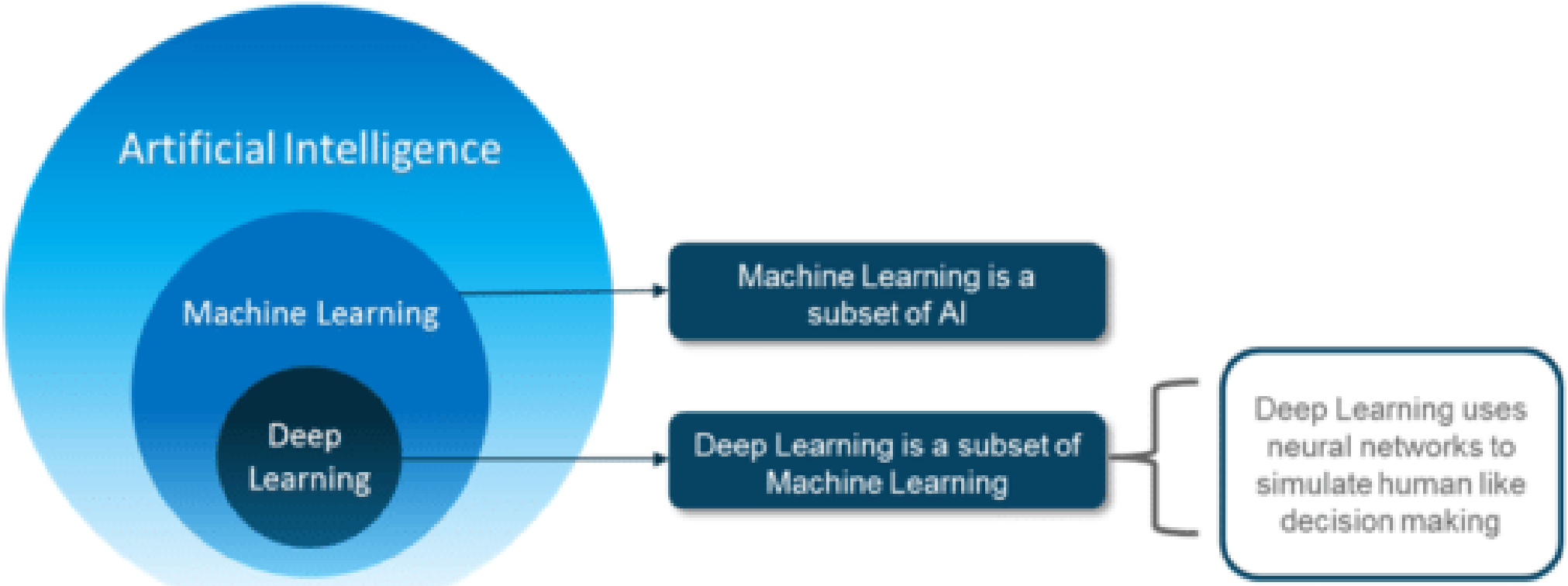
Structure of Boltzmann Machine

- During the training phase, the visible neurons are clamped; the hidden neurons always operate freely, they are used to explain underlying constraints in the environmental input vectors.
- The hidden units explain underlying constraints by capturing higher-order correlations between the clamping vectors.
- Boltzmann machine learning may be viewed as an unsupervised learning procedure for modeling a distribution that is specified by the clamping patterns.
- In Boltzmann Machine, everything is defined in terms of the energies of joint configuration of the visible and hidden units.

Applications of Boltzmann Machine

- Stock market trend prediction
- Character recognition
- Face recognition
- Internet application
- Cancer detection
- Loan application
- Decision making

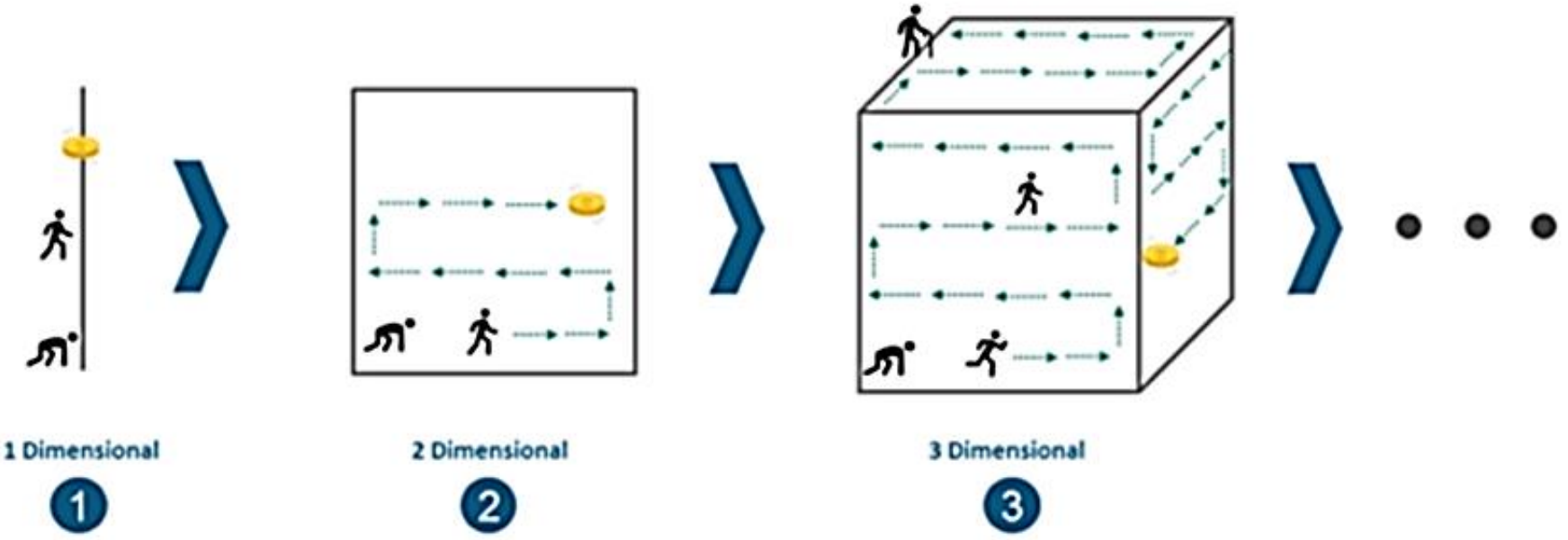
Deep Learning



Deep Learning

Machine Learning is not capable of handling high dimensional data that is where input & output is quite large. Handling and processing such type of data becomes very complex and resource exhaustive. This is termed as **Curse of Dimensionality**

Deep Learning



You have dropped a coin somewhere on the line. Now, it’s quite convenient for you to find the coin

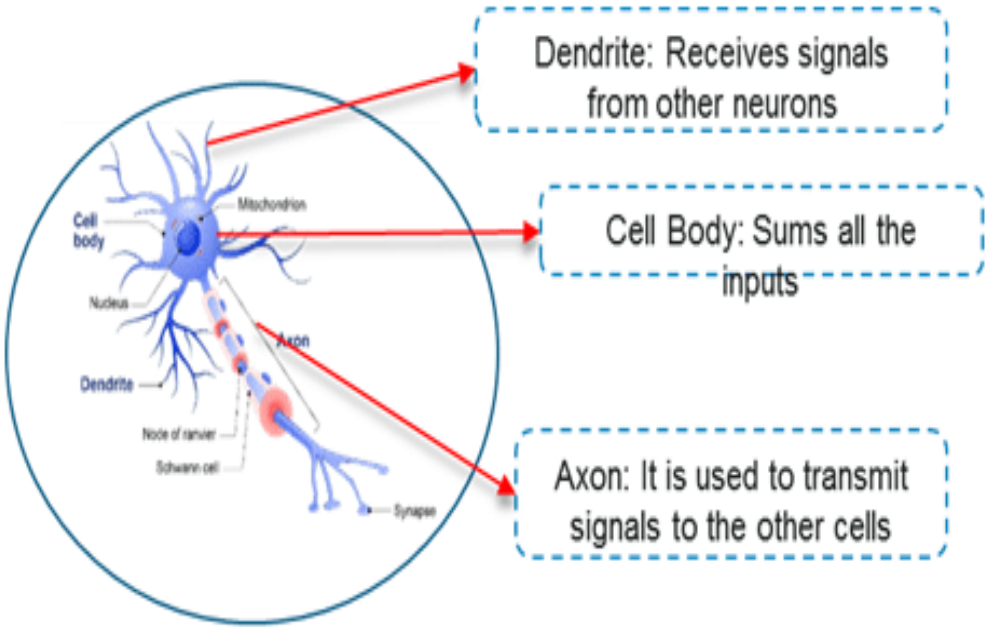
Deep Learning

Hence, we can observe the complexity is increasing as the dimensions are increasing. And in real-life, the high dimensional data that we were talking about has thousands of dimensions that makes it very complex to handle and process. The high dimensional data can easily be found in use-cases like Image processing, NLP, Image Translation etc.

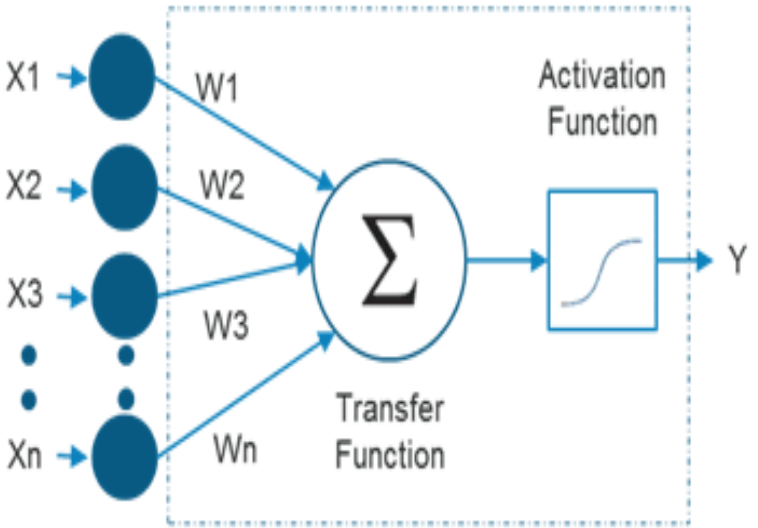
Machine learning was not capable of solving these use-cases and hence, Deep learning came to the rescue. Deep learning is capable of handling the high dimensional data and is also efficient in focusing on the right features on its own. This process is called feature extraction

How Deep Learning Works?

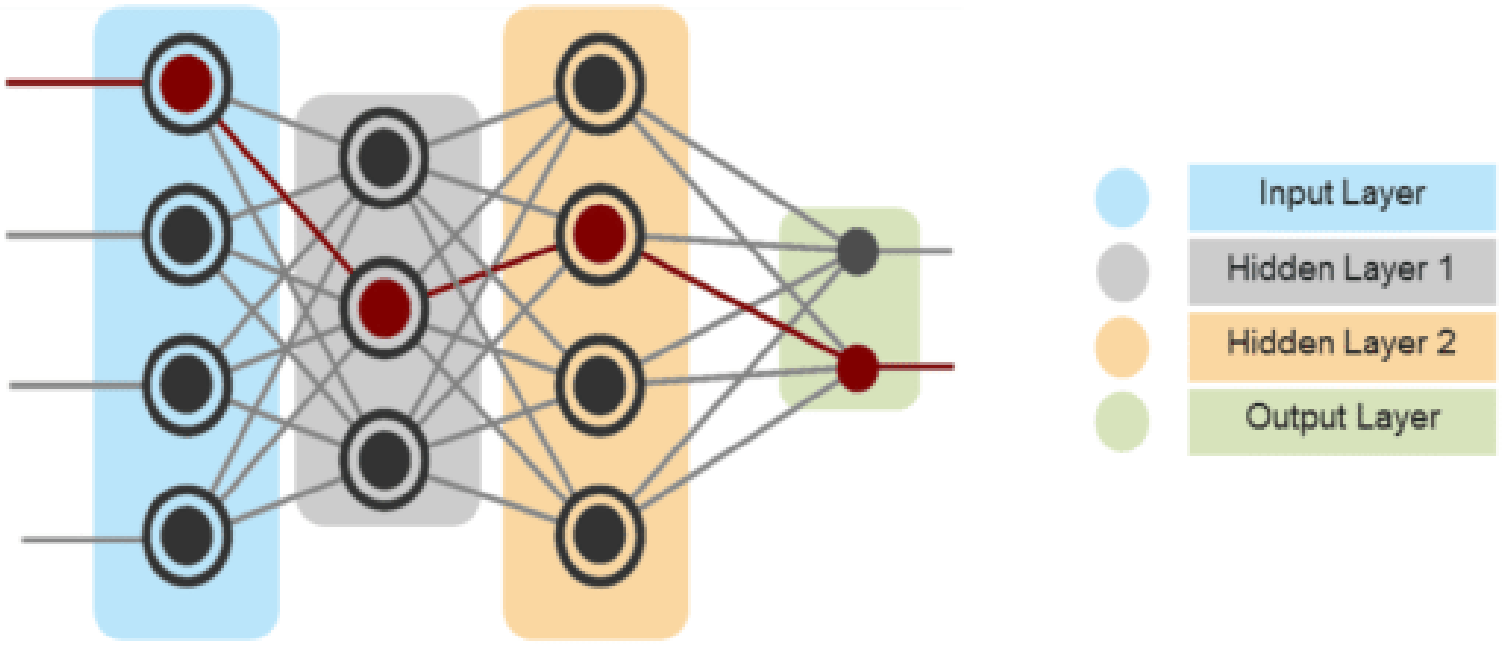
In an attempt to re-engineer a human brain,



Neuron



Schematic for a neuron in a neural net

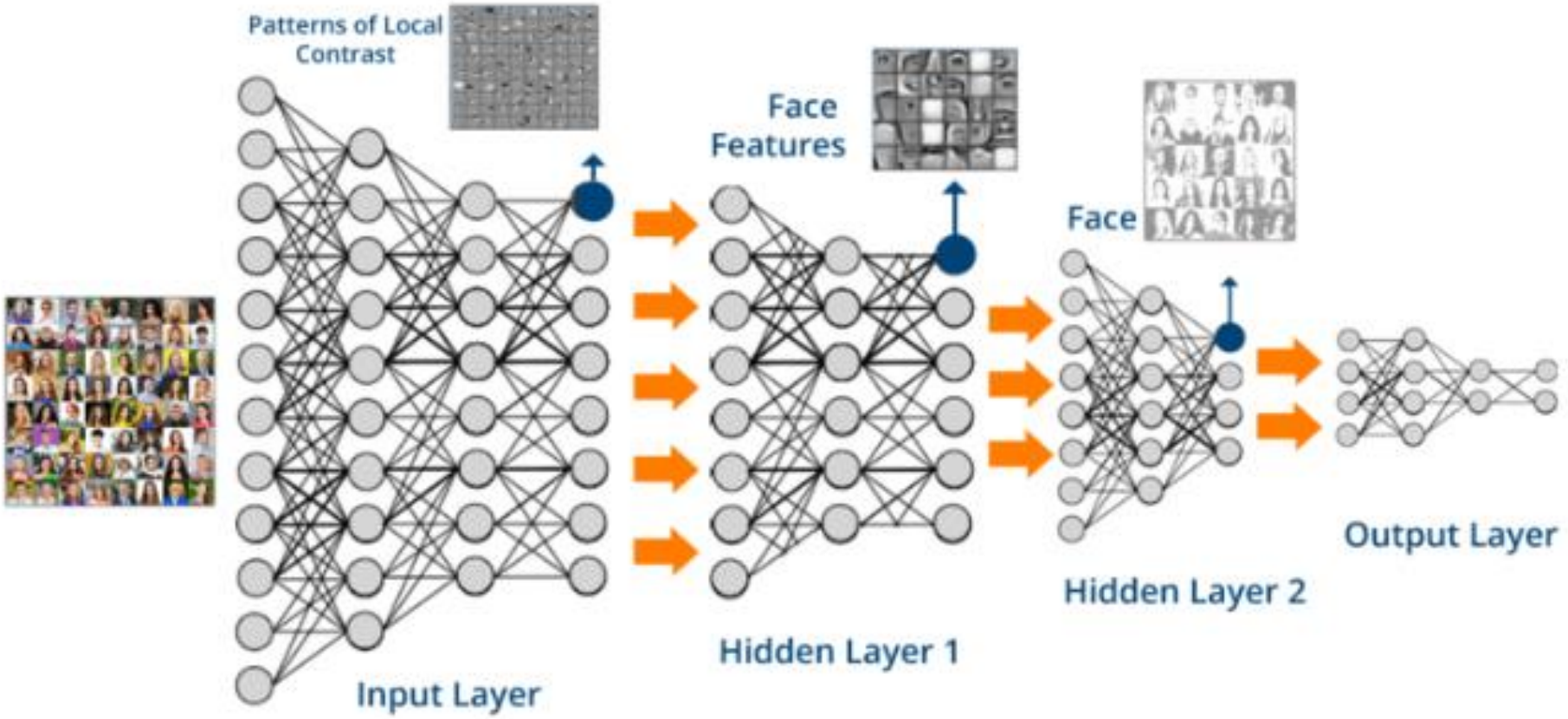


First layer is the input layer which receives all the inputs and the last layer is the output layer which provides the desired output. All the layers in between these layers are called hidden layers. There can be n number of hidden layers

Deep learning is a class of machine learning algorithms that:

- use a cascade of multiple layers for feature extraction and transformation. Each successive layer uses the output from the previous layer as input.
- learn in supervised (e.g., classification) and/or unsupervised (e.g., pattern analysis) manners.
- learn multiple levels of representations that correspond to different levels of abstraction; the levels form a hierarchy of concepts.

Suppose we want to perform Image recognition using Deep Networks



Dimensionality Reduction

In machine learning classification problems, there are often too many factors on the basis of which the final classification is done. These factors are basically variables called features. The higher the number of features, the harder it gets to visualize the training set and then work on it. Sometimes, most of these features are correlated, and hence redundant. This is where dimensionality reduction algorithms come into play. Dimensionality reduction is the process of reducing the number of random variables under consideration, by obtaining a set of principal variables. It can be divided into feature selection and feature extraction.

Components of Dimensionality Reduction

There are two components of dimensionality reduction:

- Feature selection:

  In this, we try to find a subset of the original set of variables, or features, to get a smaller subset which can be used to model the problem. It usually involves three ways:

  - **Filter** (pick up the fundamental properties of the features)
  - **Wrapper** (measure the “usefulness” of features based on the classifier performance)
  - **Embedded** (similar to wrapper but optimize the objective function or performance of a learning algorithm or model

- Feature extraction:

  This reduces the data in a high dimensional space to a lower dimension space, i.e. a space with lesser no. of dimensions.

Common Dimensionality Reduction Techniques

- ❑ Missing Value Ratio
- ❑ Low Variance Filter
- ❑ High Correlation Filter
- ❑ Random Forest
- ❑ Backward Feature Elimination
- ❑ Forward Feature Selection
- ❑ Factor Analysis
- ❑ Principal Component Analysis
- ❑ Linear Discriminant Analysis
- ❑ Independent Component Analysis
- ❑ Methods Based on Projections
- ❑ t-Distributed Stochastic Neighbor Embedding (t-SNE)
- ❑ UMAP